

How archival eclipse timing can produce period change

Three mechanisms quantified on 26 short-period binaries

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ABSTRACT

Context. A quadratic ephemeris fitted to decades of eclipse timings returns a period-change coefficient whatever the data contain: formal errors, bars from too few samples and undersampled orbits all produce one.

Aims. To quantify the three mechanisms on one uniformly processed sample and to establish what the residuals can say about the bars.

Methods. Timings of 26 binaries from TESS 2-min photometry (2019–2026) and SuperWASP (2004–2008), with bars from the data; a leverage-corrected residual test with its null simulated as the pipeline operates; injected orbits and secular terms; independent minima and a published orbit.

Results. Formal errors make 24 of 31 targets “significant”; data-derived bars leave 10 of 26 passing an adversarial gate. With bars co-estimated from two or three samples the residual test has a null of 0.77, not 1, and cannot separate correct bars from bars inflated by real terms: an injected secular term inflates its own bars, with 50% completeness at 0.015 s yr^{-1} and $\sigma(dP/dt)$ overestimated by a median factor of 2.8 at 0.2 s yr^{-1} . The ten detections are structure within five years of TESS; all but TIC 232634196 vanish under bars inflated by the secondaries’ between-sector drift, and for it quadratic and linear ephemerides predict minima 20 min apart at 2028.0.

Conclusions. Bars from two or three samples bias the residual test and absorb a real secular term. At most one or two of the ten detections are undersampled orbits. Coefficients over 16–20 yr from a few epochs are not secular rates without bars validated at the between-sector scale and an external record.

Key words. binaries: eclipsing – stars: fundamental parameters – methods: data analysis – methods: statistical – techniques: photometric – ephemerides

1. Introduction

Secular period changes in close binaries encode mass transfer, angular-momentum loss and light-travel-time effects, but measuring them requires eclipse timings spanning decades, and the archival photometry that supplies the baseline comes with heterogeneous time systems, unquantified timing biases and no ground truth internal to the archive. A quadratic ephemeris fitted to a few epochs over such a baseline returns a coefficient whatever the data contain. This paper is about the ways that coefficient can be produced, and about how to tell.

Three mechanisms are quantified on one uniformly processed sample: timing errors that are formal rather than measured; timing bars that are estimated from too few samples and applied at another time-scale, where the question of *whether* a bar is wrong is answered by a leverage-corrected decomposition of the residuals against the null of its own estimator; and third-body orbits sampled at a few phases, which a parabola absorbs into a coefficient that every internal test then certifies. The chain was developed on one system (TYC 7024-1046-1 = TIC 142874476), where a SuperWASP epoch from 2006–2008 propagated through a TESS-refined ephemeris sits 22.7 ± 2.2 min from the linear prediction, and is applied here to every catalogued binary for which the data allow the tests to run. The 26 binaries are the vehicle, not the object: they are the subset with data sufficient to demonstrate the mechanisms, drawn from

a cache rather than from the sky (Sect. 2), and no population statement is made from them.

None of the three mechanisms is unknown. That formal photon-noise errors underestimate eclipse-timing uncertainties is why timing studies use the scatter of the data; that a light-travel-time orbit sampled over a fraction of its period is degenerate with a quadratic ephemeris is a standing caveat of the eclipse-timing literature, stated wherever a cyclic and a secular term are fitted to the same O–C (e.g. Borkovits et al. 2016; Hajdu et al. 2019), and the timing of eclipsing binaries from TESS itself has been examined for exactly this degeneracy (Marcadon & Prša 2024). What is new here is not the mechanisms but their measurement on one sample with one chain: a detection efficiency for undersampled orbits measured target by target at the real epochs with the real bars; one target with a published orbit and no secular term (V527 Dra) in which every internal test — significance, adversarial, goodness of fit — passes the confounder; and a leverage-corrected decomposition of the residuals, with the null distribution of the estimator obtained by re-estimating the bars inside the simulation, which says whether either archive carries excess variance and — in this sample — traces an apparent excess of 2.73 to two estimation errors rather than to the data.

Table 1. Selection funnel.

Step	Criterion	N
1	Union of the TESS eclipsing-binary catalogues of Prša et al. (2022; 4 584) and Kostov et al. (2025; 7 936), deduplicated by position (5'') over the merged table	12 497
2	Catalogue period $P < 5$ d	8 759
3	≥ 2 TESS 2-min sectors separated by ≥ 1 yr among the 22 sector manifests held locally (s19–21, 27–28, 75, 90–105)	288
4	A SuperWASP source within 20''(public archive cone search — the query that lists sources, not the light-curve download; 0 of 288 queries unresolved after retry)	131
5	≥ 4 independent epoch clusters (TESS sectors + SuperWASP seasons), the minimum for a quadratic ephemeris with ≥ 1 degree of freedom	88
6	Passed all measurement guards (Sect. 3)	26

2. The sample as a vehicle: 26 binaries with data sufficient for the tests

Selection is deterministic and every cut is listed with its survivors (Table 1). The sample exists to run the tests of Sect. 5, not to represent a population; the targets that were *not* measured are nevertheless reported with the reason (Table 2), because the guards that stopped them are part of the method.

Two remarks. On step 1: the two catalogues are disjoint by construction (0 TICs in common; 23 positional pairs at 5''), so the union genuinely adds.

On step 3, a caveat that anyone reproducing the selection needs: **the sector count is a measure of our cache, not of the sky**. The 22 local manifests cover 21% of TESS sectors, concentrated in 2019–2020 and 2024–2026, so 288 is a lower bound — and a *biased* one, because the undercount is not uniform.

The cache census. The bias was measured rather than left as a caveat (`src/vies_cache_288.py`; expectation written before running: a median ratio of 2–3 and 15–25% of targets at a ratio ≥ 5). For each of the 288 targets that passed step 3, MAST was asked how many SPOC 2-min sectors exist (288 of 288 answered, 0 network failures). The cache holds a median of 4 sectors per target; the archive holds a median of 32 (2–45), 7 857 sectors in all against 1 224 cached. The ratio archive/cache has median 5.7 (quartiles 4.0–10.3, 10th–90th percentiles 2.3–12.0), is ≥ 5 for 62% of the 288 and equal to 1 for 1%, and rises with ecliptic latitude (Spearman +0.60, $p = 5 \times 10^{-30}$): median 2.3 at $|\beta| < 40^\circ$ ($n = 29$, none ≥ 5), 2.8 at 40 – 60° (36), 6.0 at 60 – 78° (82) and 10.0 at 78 – 90° (141, 83% ≥ 5). Half of the 288 sit in the continuous-viewing zones, where the cache holds a tenth of what exists. The 26 measured targets are the extreme of this: cache median 4, archive median 39, ratio 10.5. The expectation was wrong by a factor of two in the median and three in the fraction above 5: the undercount is severe everywhere and worst exactly where the sample lives.

What this implies is stated once and used throughout: this is not a sample of short-period binaries, it is a sample of binaries in the continuous-viewing zones with deep archival photometry — which is what makes it adequate as a vehicle for the tests of Sect. 5 and inadequate as a population. The sky distribution of

the result follows directly: 21 of the 26 measured targets are in Draco (the others: 2 in Microscopium, 2 in Grus, 1 in Indus), because the cached northern sectors overlap there and SuperWASP covered it in 2004–2008. The sample is a Draco sample with a southern tail, not an all-sky one — which is why no population claim is made from it (Sect. 6).

Why the cached sectors and not the archive. The chain was run on the sectors already in the local cache — those downloaded for the earlier searches, a median of four per target — and not on the 799 2-min light curves that MAST holds for the 26, although a manifest of them was built and is kept with the code. Three reasons, in the order they weighed. The tests of Sect. 5 are about what the chain does with the data it is given, and four sectors per target is the regime in which the archival seasons carry the coefficient — the regime this paper is about; the control target, the comparison between the versions of the chain and every expectation written here are for that design, and changing the design in the same step as correcting the estimator would have hidden which of the two moved the numbers. And the full archive changes the question rather than the bars: with 30–45 sectors over 2019–2025 the TESS block alone constrains curvature over six years, the period ladder — designed and tested on 4–7 epochs — has to be rebuilt across sectors of different noise, and the between-sector drift measured on five secondaries (Sect. 5.3) becomes measurable on every target. What would change is therefore stated as an expectation, written and committed with the manifest: the mean TESS leverage falls from 0.62 to about 0.1 and the residual test gains power on the TESS side, where a variance factor of 1.5–3 is expected from the drift the secondaries show; the ten detections carried by four sectors are expected to be either confirmed as continuous six-year structure or dissolved into between-sector drift, with 3–6 surviving, TIC 232634196 among them; no new detection beyond 2 of 26. That run is the next version of this work, not a variant of this one.

A recovery pass was run on the 8 + 6 “recoverable” rows after the main batch (intermediate TESS sectors from MAST for the eight; eclipse durations measured on the TESS curves for the six). It recovered none of them: Table 2 lists the *first* guard each target failed, and behind the first there was a second in every case. Recovery is therefore not additive across the reasons in Table 2, and the count of 26 stands (the per-target outcome is in Appendix A).

The 26 measured targets are therefore not a random sample of short-period binaries: they are the subset with deep SuperWASP coverage, eclipses sampled in every season, and unambiguous cycle counts across the TESS sectors held locally.

3. Method

3.1. Eclipse epochs

For each TESS sector we fit a periodic trapezoid of free depth to the PDCSAP light curve (QUALITY = 0, normalised to the 95th percentile) on a fine grid of trial epochs, with the eclipse duration T_{14} from the catalogue and the period from the catalogue, seeded on the catalogue reference epoch so that the primary is fitted rather than the deepest minimum in the curve (in contact systems these differ). The epoch is the χ^2 minimum; the formal error is the $\Delta\chi^2 = 1$ half-width. Sectors are anchored to the most recent one.

For SuperWASP we convert HJD_UTC to BJD_TDB (a +1.1 min shift at these dates — larger than the TESS timing error), fit the same profile per observing season, and require phase coverage of the eclipse in every season (≥ 50 points in eclipse, no

Table 2. The 62 targets of the 88 that were not measured, by the first guard each one failed. This table is a record of where each target stopped, not a map of what recovery would gain: every target fails at the first guard it meets, and behind the first there is usually another (see the recovery pass below). The “Nature” column describes the first reason only.

Reason	N	Nature
SuperWASP curve with < 500 valid points after cleaning	21	archive depth; structural
Phase coverage of the eclipse not testable in ≥ 1 season block (Sect. 3.3)	22	archive sampling; recoverable with a second archive
SuperWASP light-curve download returned an empty file (a different operation from the cone search of Table 1, step 4)	4	fetch failure without retry in this run; recoverable (retry now implemented)
Cycle count between TESS sectors ambiguous (period ladder did not close)	8	local sector coverage; intermediate sectors were added from MAST and none was recovered (Appendix A)
No applicable eclipse duration in the catalogue (4 NaN; 1 contact-binary width; 1 W UMa)	6	catalogue gap; T_{14} was measured on the TESS curve for five and none was recovered (Appendix A)
Catalogue period error NaN	1	catalogue gap

empty phase bins in 12). A season whose fitted depth is ≤ 0 is rejected: the grid has found a χ^2 minimum for a *brightening*, and its “epoch” is not that of an eclipse. Five of the first 31 targets had such a season; their between-season dispersion was 82 min against 1.8 min for the rest.

Blending. A constant neighbour in the aperture dilutes the depth of an eclipse and does not move its time of minimum; only a neighbour that is itself variable on the time-scale of the eclipse can shift a minimum or add timing structure. What can be measured cheaply is who has a neighbour capable of it (`src/blending_26.py`; expectation written and committed before running). The SPOC crowding metric CROWDSAP of the 84 sectors used has a median of 0.97 per target (median over its sectors), 14 of the 84 sectors are below 0.9 and none below 0.5; the lowest is TIC 377253090 (0.56 in all four sectors). Gaia DR3 sources within 48'' (about two TESS pixels) and within 2 mag in G of the target exist for three of the 26: TIC 229476285 (18.8'', $\Delta G = 1.5$), TIC 359552377 (10.1'', $\Delta G = 1.9$), and TIC 377253090, a member of D10, which has a companion of nearly equal brightness at 1.7'' ($\Delta G = 0.37$) that neither TESS nor SuperWASP resolves: which of the two stars is the eclipsing binary, and whether the other is variable, is not decided by the data of this paper, and the detection is carried with that caveat. No blending correction is applied to any epoch, since none is needed for a constant neighbour; the variability of the neighbours was not tested.

3.2. Period ladder and cycle count

Within the TESS sectors the period is refined by a ladder: starting from the two closest sectors, each further sector is added only if its linear residual stays below a tolerance equal to the larger of 3 min and three times the median epoch uncertainty of the target (3 min for 22 of the 26; 4.1–5.5 min for the other four), and the result is discarded if shifting any epoch by ± 1 cycle does not worsen χ^2 . The ladder returns epochs and residuals in the caller’s order (Sect. 5).

The ladder is *not* used across the 2008–2019 gap: it requires a linear ephemeris, and a real dP/dt produces tens of minutes of O–C over 19 yr, which the ladder would reject as an unresolved alias. Across the gap the cycle number is the rounded propagation with the TESS-refined period, and the count is accepted only if $\sigma_P \times N < 0.25$ cycle at the farthest epoch. Any target with $|dP/dt| > 10 \text{ s yr}^{-1}$ is flagged as a suspected cycle-count error rather than a measurement; none of the 26 is.

3.3. Timing uncertainties from the data

The formal $\Delta\chi^2 = 1$ error is a photon-noise error. Cycle-to-cycle eclipse-timing scatter from spots and O’Connell-type asymmetries is not in it, and in the SuperWASP data the dispersion between observing seasons exceeds the formal error by a median factor of 4.3 ($n = 25$ targets of the 26; one record lacks the formal value; interquartile range of the factor 2.9–10.7). We therefore assign each epoch the *larger* of its formal error and the dispersion between independent subsets: the two halves of the sector for TESS, the observing seasons for SuperWASP. Each SuperWASP season enters the fit as its own epoch with this error — the dispersion of one season, $\text{std}(d)$, not the standard error of the season mean, $\text{std}(d)/\sqrt{n}$, which is the bar of the *combined* epoch used on the control target; an earlier version assigned the latter to each season (Sect. 5.2.2, Appendix D) — added in quadrature to the chain-bias uncertainty of Sect. 3.4. A target with a single valid season has no red-noise estimate and is not measured. Two properties of these bars are shown by the data: because each is estimated from two or three samples that also enter the residual, the residual test has a null below 1 that must be simulated rather than assumed (Sect. 5.2.3), and the TESS bar, being a within-sector quantity, cannot contain timing drift between sectors (Sect. 5.3.2).

3.4. SuperWASP timing bias

There is no ground truth internal to the archive, so the timing bias of the SuperWASP chain was calibrated on transiting hot Jupiters — objects whose clocks do not run — with TESS ephemerides precise enough to propagate 18 yr. Eclipsing binaries cannot serve as clock calibrators: a first attempt with three gave an astrophysical O–C scatter of 31.7 min, against 1.73 min for the hot Jupiters. Of 12 hot Jupiters, 8 yielded epochs and 4 passed the between-season dispersion guard (WASP-18 b, WASP-4 b, WASP-6 b, WASP-19 b); their O–C are -2.85 ± 1.07 , -2.55 ± 1.52 , $+0.85 \pm 2.21$ and -0.64 ± 2.85 min, consistent with a constant offset ($\chi^2 = 2.62$ for 3 d.o.f.):

$$\text{chain bias} = -2.14 \pm 0.78 \text{ min.} \quad (1)$$

The bias is subtracted from every SuperWASP epoch and its uncertainty added in quadrature. The four rejected calibrators (between-season bars of 17–951 min) were not discarded silently; the guard that rejected them is the same one applied to every target here.

Declared limitation: the ± 0.78 min is a floor, not an estimate of the bias uncertainty for the eclipsing binaries, and the transfer of the bias itself is asserted, not demonstrated.

A bias of -2.14 min is too large for a time-scale error (the HJD $\rightarrow$ BJD and UTC $\rightarrow$ TDB conversions are already applied and amount to ~ 1.1 min in total), so it is procedural — it arises in the fitting, the sampling or the profile — and a procedural bias is shape-dependent. It was measured on shallow, long transits (depth $\sim 1\%$, duration 2–3 h) and is transferred here to deep, sharp eclipses of 1–2 h, without an argument that the two shapes bias the same way. The four surviving calibrators are, moreover, planets discovered by WASP itself, i.e. the objects with the best coverage in the archive. The only eclipsing-binary check available is the data-level overlay of Sect. 5.5 (two SuperWASP-era epochs of CV Dra against independent CCD minima, differences -0.3 ± 2.7 and $+3.6 \pm 2.3$ min): consistent with the transferred bias, and far from a measurement of it. The uncertainty of every SuperWASP epoch therefore carries a term whose true size is unknown and is at least 0.78 min. There is no cheap way to resolve this, and the paper does not try: the calibration on four hot Jupiters is carried as a declared limitation, and the only measurement that bounds it is the data-level agreement of two SuperWASP-era epochs of CV Dra with independent CCD minima ($n = 2$). The SuperWASP residuals of the 23 targets with an adequate quadratic do not bound it, and an earlier version of this paragraph said they did: a shape-dependent bias is, to first order, the same for every season of a target, i.e. a common offset of the archival block against the TESS block, and a common offset is absorbed by the quadratic — it is the shape most degenerate with the curvature — leaving no residual. The residual test of Sect. 5.2.3 is blind to exactly the term it was being used to limit; what it does bound is a season-to-season scatter beyond the bars, which is a different thing. One more thing the ± 0.78 min is not: independent from one season to the next. Added in quadrature to each epoch, as the tables do, it is averaged down by $\sqrt{n}$ over a target's seasons, whereas the uncertainty of a chain bias is a common offset of the whole archival block against the TESS block — the shape most degenerate with curvature. The fits were therefore repeated by generalised least squares with the 0.78^2 term moved from the diagonal into a common block over each target's SuperWASP epochs (src/gls_vies_26.py; expectation written and committed before running: no change of verdict in the nine detections carried by TESS, a wider σ for the one carried by the archive). No verdict changes in any of the 26 and no new detection appears; $\sigma(dP/dt)$ grows by a median of 2% and by at most 16% (StKM 1-1676, TIC 230386284, whose bars are the smallest of the sample), and — against the expectation — not at all for TIC 232634196, whose 19.8-min season bars are 25 times the bias term, so that its placement is immaterial. The tables keep the diagonal treatment; the correlated one is the correct propagation and changes nothing that is claimed.

3.5. Fit and tests

Linear and quadratic ephemerides are fitted by weighted least squares to all epochs (TESS sectors + SuperWASP seasons). $dP/dt = 2Q/P$ with Q the quadratic coefficient. Degrees of freedom are $N - 3$; a target with $N < 4$ would be “measured, not tested” (none of the 26 is). Curvature significance is $\Delta\chi^2$ between the two fits with one degree of freedom.

Three tests accompany every number. The *goodness of fit of the quadratic ephemeris* is the χ^2 of the fit with its $N - 3$ degrees of freedom, reported as p_{gof} ; a fit with $p_{\text{gof}} < 0.05$ is flagged “parabola inadequate” and its coefficient is not quoted

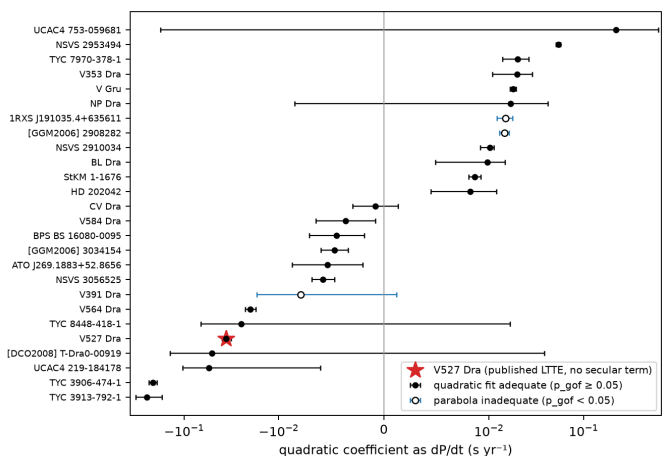


Fig. 1. Quadratic coefficients dP/dt of the 26 targets with 1σ bars, ordered by value; the three flagged targets ($p_{\text{gof}} < 0.05$) are shown open, and V527 Dra is marked.

as a rate. The level is the same as that of the curvature test, chosen for consistency and not for what it marks; an earlier draft used a fixed $\chi^2_{\text{red}} \geq 2$, which corresponds to a false-flag probability that depends on the degrees of freedom (0.157 at 1 d.o.f., 0.135 at 2, 0.112 at 3, 0.092 at 4) — the counts under both rules are reported in Sect. 4. The *curvature test* is the $\Delta\chi^2$ above. The *adversarial test* shifts every epoch by one sigma in the direction that most weakens the curvature and reports the resulting p-value; curvature is called only if $p < 0.05$ both before and after. The shifted value is not a p-value — a coherent one-sigma shift of every epoch has no null distribution — and the test is a robustness heuristic of unquantified conservatism, used here as a gate and not as a significance level. Sect. 5 shows what it does and does not protect against.

4. The chain applied to the 26

This section records what the chain returns with the data-derived bars of Sect. 3.3, before any of the tests of Sect. 5. Table 3 is the per-target record on which Sect. 5 operates; Table 4 summarises it in three partitions. Neither is a population result: the coefficients are quadratic coefficients over 16–20 yr, and Sect. 5 shows what they survive.

Five remarks on the table (Fig. 1 shows the coefficients; Fig. 3 shows what the detections track). (i) In every partition the distribution is consistent with symmetry about zero: 12/14 (sign test $p = 0.85$), 10/13 ($p = 0.68$) and 8/8 ($p = 1.00$). (ii) The median $|dP/dt|$ of 0.016 s yr^{-1} over all 26 is $1.8 \times 10^{-7} \text{ d yr}^{-1}$, the order of magnitude familiar from period studies of W UMa systems; 0.009 s yr^{-1} for the well-measured 16 is $1.1 \times 10^{-7} \text{ d yr}^{-1}$. That median coincides with the amplitude at which the chain's completeness for a secular term is 50% (0.015 s yr^{-1} ; Sect. 5.2.5). The 26 include the non-detections, so the distribution is not a truncated one, but it is shaped by the sensitivity of the chain at exactly the scale it reports, and the agreement with the W UMa literature — itself built from chains with their own thresholds — is not read here as an astrophysical result. (iii) Under the earlier fixed rule $\chi^2_{\text{red}} \geq 2$ the flag would have marked 5 targets instead of 3 (adding TIC 207496824 and 424461577, whose p_{gof} are 0.12 and 0.11); with 5 of 26 flagged, the chance that such a flag lands on one given target by coincidence is 19% (27% with the 7 of 26 the earlier bars flagged), a number that matters in Sect. 5.4. (iv) Three targets have curvature at $0.01 <$

Table 3. Quadratic ephemerides of the 26 measured targets. Name is the variable-star designation where one exists, otherwise the SIMBAD identifier; positions are those of the TIC (J2000, degrees); T is the TESS magnitude; P the catalogue period in days; N the number of epochs; $\nu = N - 3$; span the baseline in years; dP/dr the quadratic coefficient expressed as a rate, in s yr^{-1} . Class: C, curvature surviving the adversarial test; –, none; a, fails the adversarial test; †, parabola inadequate ($P_{\text{gof}} < 0.05$), the coefficient then being that of a model that does not describe the data and listed for completeness only. No entry of this table is a period-change rate: each is a quadratic coefficient over 16–20 yr, and Sect. 5 shows what such a coefficient can be made of; the table is not to be cited as measured dP/dr (Sect. 6). V527 Dra (TIC 424461577) passes all three tests and has a published cyclic O–C (Sect. 5.4).

TIC	Name	RA	Dec	T	P	N	ν	span	dP/dr	χ^2_{red}	P_{gof}	P_{curv}	P_{adv}	cl.
115244268	TYC 7970-378-1	317.6428	-40.5239	10.9	2.0089	5 (3+2)	2	19.9	+0.0200 ± 0.0061	0.44	0.645	0.001	0.286	–a
126763885	UCAC4 219-184178	316.2145	-46.3262	12.4	4.3574	4 (2+2)	1	19.9	-0.0549 ± 0.0489	1.23	0.268	0.261	0.485	–
126945917	HD 202042	318.7324	-43.3954	9.1	1.3411	6 (4+2)	3	19.9	+0.0082 ± 0.0037	0.44	0.722	0.028	0.805	–a
139256217	TYC 8448-418-1	347.1903	-46.1102	11.0	2.1992	4 (2+2)	1	19.9	-0.0249 ± 0.0417	1.09	0.297	0.550	0.987	–
144194304	V Gru	327.9727	-42.3732	9.2	0.4835	5 (2+3)	2	19.9	+0.0181 ± 0.0014	0.58	0.560	< 10 ⁻³	< 10 ⁻³	C
198388252	[GGM2006] 2908282	258.1953	+57.0529	11.6	0.3952	7 (4+3)	4	19.6	+0.0146 ± 0.0017	5.31	< 10 ⁻³	< 10 ⁻³	< 10 ⁻³	C†
198408416	NSVS 2910034	258.9133	+58.8645	12.1	0.3647	7 (4+3)	4	19.5	+0.0102 ± 0.0011	0.70	0.594	< 10 ⁻³	< 10 ⁻³	C
199688409	BPS BS 16080-0095	253.0515	+57.7255	12.0	0.8683	7 (4+3)	4	17.6	-0.0045 ± 0.0026	0.23	0.922	0.087	0.795	–
207496824	V353 Dra	246.9546	+58.8398	10.8	0.8027	4 (2+2)	1	16.7	+0.0197 ± 0.0089	2.45	0.117	0.027	0.593	–a
224605072	[DCO2008] T-Dra0-00919	255.5992	+58.8700	11.4	0.6649	7 (4+3)	4	17.6	-0.0510 ± 0.0893	0.64	0.635	0.568	0.983	–
229461186	BL Dra	295.1026	+60.9203	10.9	0.4027	5 (3+2)	2	15.7	+0.0099 ± 0.0049	0.72	0.487	0.046	0.527	–a
229476285	UCAC4 753-059681	295.5345	+60.4830	12.9	1.0965	5 (3+2)	2	15.7	+0.2200 ± 0.3975	0.79	0.454	0.580	0.765	–
229687624	CV Dra	262.9885	+57.1784	9.0	0.6176	7 (4+3)	4	19.6	-0.0008 ± 0.0022	0.57	0.683	0.713	0.936	–
229914020	IRXS J191035.4+635611	287.6436	+63.9373	11.9	0.6760	5 (3+2)	2	15.7	+0.0149 ± 0.0028	3.05	0.047	< 10 ⁻³	< 10 ⁻³	C†
230386284	StKM 1-1676	285.8220	+63.9928	9.0	0.3415	5 (3+2)	2	15.7	+0.0087 ± 0.0006	0.07	0.935	< 10 ⁻³	< 10 ⁻³	C
232634196	TYC 3906-474-1	269.4385	+56.1960	9.9	1.2511	7 (4+3)	4	19.6	-0.2141 ± 0.0208	0.60	0.660	< 10 ⁻³	< 10 ⁻³	C
237116051	V391 Dra	269.7931	+58.7165	10.2	2.4629	4 (2+2)	1	19.6	-0.0079 ± 0.0091	13.98	< 10 ⁻³	0.387	0.722	–†
243352373	V584 Dra	290.2686	+56.3283	10.2	0.3343	4 (2+2)	1	16.7	-0.0036 ± 0.0028	0.77	0.379	0.196	0.885	–
329246824	V564 Dra	263.8847	+57.8025	11.0	0.5877	7 (4+3)	4	19.6	-0.0200 ± 0.0026	0.17	0.956	< 10 ⁻³	< 10 ⁻³	C
329248002	NP Dra	263.8179	+55.0034	8.8	3.1089	6 (4+2)	3	16.7	+0.0169 ± 0.0253	0.83	0.477	0.506	0.711	–
329289186	ATO J269.1883+52.8656	269.1883	+52.8657	13.8	0.5525	5 (2+3)	2	19.6	-0.0054 ± 0.0034	0.60	0.547	0.111	0.919	–
359552377	TYC 3913-792-1	279.3983	+57.7443	10.9	3.4391	5 (3+2)	2	16.7	-0.2485 ± 0.0769	0.48	0.617	0.001	0.219	–a
377253090	NSVS 3056525	286.1923	+58.9513	11.8	0.4245	6 (4+2)	3	16.7	-0.0058 ± 0.0011	0.28	0.838	< 10 ⁻³	0.002	C
390021728	[GGM2006] 3034154	277.7934	+56.4175	11.5	0.3020	6 (4+2)	3	16.6	-0.0047 ± 0.0013	0.93	0.427	< 10 ⁻³	0.099	–a
392536812	NSVS 2953494	280.4674	+58.0565	11.2	0.5910	6 (4+2)	3	16.7	+0.0541 ± 0.0029	0.25	0.861	< 10 ⁻³	< 10 ⁻³	C
424461577	V527 Dra	279.7343	+60.3982	10.2	0.7448	6 (4+2)	3	15.7	-0.0359 ± 0.0040	2.03	0.107	< 10 ⁻³	< 10 ⁻³	C

Table 4. The distribution, in three partitions: all 26 coefficients; the 23 whose quadratic fit is adequate ($p_{\text{gof}} \geq 0.05$); and the 16 that are adequate and have $\sigma(dP/dt) < 0.01 \text{ s yr}^{-1}$, a cut on precision rather than on value. No column is a headline: the median $|dP/dt|$ moves from 0.009 to 0.018 s yr^{-1} between them, and that sensitivity is the finding.

Quantity	n = 26 (all)	n = 23 (quadratic fit adequate, $p_{\text{gof}} \geq 0.05$)	n = 16 (adequate and $\sigma(dP/dt) < 0.01 \text{ s yr}^{-1}$)
dP/dt , median	-0.0022 s yr^{-1}	-0.0036 s yr^{-1}	+0.0037 s yr^{-1}
$ dP/dt $, median	0.0159 s yr^{-1}	0.0181 s yr^{-1}	0.0093 s yr^{-1}
dP/dt , interquartile range	[-0.0170, +0.0148] s yr^{-1}	[-0.0224, +0.0136] s yr^{-1}	[-0.0049, +0.0122] s yr^{-1}
dP/dt , minimum and maximum	[-0.248, +0.220] s yr^{-1}	[-0.248, +0.220] s yr^{-1}	[-0.036, +0.054] s yr^{-1}
Sign of dP/dt : positive / negative (two-sided sign test p)	12 / 14 (p = 0.85)	10 / 13 (p = 0.68)	8 / 8 (p = 1.00)
$\sigma(dP/dt)$, median	0.0036 s yr^{-1}	0.0037 s yr^{-1}	0.0027 s yr^{-1}
χ^2_{red} of quadratic fit, median (25th–75th)	0.67 (0.45–1.05)	0.60 (0.44–0.81)	0.58 (0.27–0.73)
$ dP/dt /\sigma > 2 / > 3 / > 5$	15 / 13 / 10	13 / 11 / 8	11 / 9 / 7
Curvature with $p < 0.05$	16	14	12
Curvature surviving the adversarial test	10 (38%, 95% CI 22%–57%) — 6 > 0, 4 < 0	8 (35%, 95% CI 19%–55%) — 4 > 0, 4 < 0	7 (44%, 95% CI 23%–67%) — 4 > 0, 3 < 0
Same, over targets with detection efficiency $\varepsilon > 0$ (Sect. 5.4)	10 of 24 (42%, 95% CI 24%–61%)	8 of 21 (38%, 95% CI 21%–59%)	—
Same, with TESS bars inflated by 1.0 / 2.6 min yr^{-1} (Sect. 5.3, sensitivity ceiling)	1 / 0 of 26	1 / 0 of 23	0 / 0 of 16
Measured but not tested (d.o.f. < 1)	0	0	0
$ dP/dt > 10 \text{ s yr}^{-1}$ (cycle-count guard)	0	0	0

Table 5. Curvature with the TESS bars inflated in quadrature by 1.0 and by 2.6 min yr^{-1} times the target lever, the sensitivity ceiling of Sect. 5.3.2, as p_{curv} , p_{adv} ; the SuperWASP bars are those of Sect. 3.3, uncorrected for the reason given in Sect. 5.2.3. A detection requires both $p < 0.05$ (marked *).

TIC	Name	TESS +1.0 min yr^{-1}		TESS +2.6 min yr^{-1}	
		p_{curv}	p_{adv}	p_{curv}	p_{adv}
115244268	TYC 7970-378-1	0.523	0.595	0.516	0.771
126763885	UCAC4 219-184178	0.791	0.955	0.852	0.944
126945917	HD 202042	0.581	0.799	0.518	0.771
139256217	TYC 8448-418-1	0.786	0.879	0.994	0.994
144194304	V Gru	0.265	0.900	0.681	0.703
198388252	[GGM2006] 2908282	0.781	0.904	0.123	0.636
198408416	NSVS 2910034	0.480	0.599	0.991	0.997
199688409	BPS BS 16080-0095	0.654	0.953	0.635	0.649
207496824	V353 Dra	0.847	0.923	0.484	0.739
224605072	[DCO2008] T-Dra0-00919	0.457	0.919	0.485	0.834
229461186	BL Dra	0.438	0.592	0.668	0.871
229476285	UCAC4 753-059681	0.371	0.719	0.375	0.819
229687624	CV Dra	0.454	0.665	0.436	0.721
229914020	1RXS J191035.4+635611	0.515	0.771	0.776	0.913
230386284	StKM 1-1676	0.372	0.597	0.680	0.916
232634196	TYC 3906-474-1	$< 10^{-3}$	$< 10^{-3}$ *	0.012	0.745
237116051	V391 Dra	0.001	0.174	0.001	0.166
243352373	V584 Dra	0.636	0.703	0.605	0.650
329246824	V564 Dra	0.168	0.639	0.561	0.612
329248002	NP Dra	0.736	0.748	0.697	0.799
329289186	ATO J269.1883+52.8656	0.436	0.906	0.406	0.763
359552377	TYC 3913-792-1	0.069	0.725	0.348	0.722
377253090	NSVS 3056525	0.481	0.652	0.586	0.656
390021728	[GGM2006] 3034154	0.903	0.913	0.495	0.628
392536812	NSVS 2953494	$< 10^{-3}$	0.218	0.248	0.548
424461577	V527 Dra	0.078	0.831	0.443	0.575

$p_{\text{curv}} < 0.05$ (TIC 126945917, 207496824, 229461186); with 26 targets tested at the 0.05 level about one such marginal value is expected by chance, and all three fail the adversarial test in any case, so a look-elsewhere correction changes no count. (v) Two targets (TIC 224605072, 229476285) have no detection sensitivity at the amplitudes considered in Sect. 5.4 ($\varepsilon = 0$: their SuperWASP bars are too large for the curvature test to reject anything), so their “no significant curvature” is not a null result and they do not belong in the denominator of a curvature fraction; over the 24 with sensitivity the fraction is 10 of 24 (42%, 95% CI 24–61%), given as a separate row of Table 4. The last row of Table 4 and Table 5 give the detections that survive when the TESS bars are

inflated by the between-sector drift (Sect. 5.3.2); no SuperWASP correction is applied, for the reason given in Sect. 5.2.3.

Two sets are used by name in the rest of the paper. **D10** is the ten targets whose curvature survives the adversarial test (TIC 144194304, 198388252, 198408416, 229914020, 230386284, 232634196, 329246824, 377253090, 392536812, 424461577); **D8** is D10 without the two flagged members (198388252, 229914020). One member of D8, TIC 377253090, has an unresolved companion of equal brightness at $1.7''$ (CROWDSAP 0.56; Sect. 3.1) and is carried with that caveat. The completeness of the chain for a secular term, measured by injection with the same estimator (Sect. 5.2.5), is 50% at 0.015 s yr^{-1} and, at the

tabulated coefficients of the ten, 0.54–1.00: the ten are a count, not a rate. The third flagged target, TIC 237116051, has no surviving curvature. For the two flagged members of D10 the reading is *period variation detected without a determined form*; for the third, *not described by a quadratic ephemeris*; in neither case should the tabulated coefficient be quoted as a rate. The third column of Table 4 (adequate fit and $\sigma < 0.01 \text{ s yr}^{-1}$) contains 7 members of D8; it is a precision cut, not a third set. An earlier version had eleven and nine (Appendix D): TIC 359552377 left the set when the composition error of Sect. 5.2.2 was corrected.

5. Three mechanisms that can produce a period change, and what the data can test

Each mechanism is stated, quantified on the 26, and confronted with what the data can and cannot test. The demonstration that frames all three is external and independent of the cache: for CV Dra, the one target with 34 years of independent CCD minima, a quadratic fitted to the same 2004–2024 window reproduces our coefficient (4 of 4 agree), and a quadratic fitted to the whole record does not (2 of 5; Fig. 2; Sect. 5.5). The object called “dP/dt over 16–20 yr” is a local coefficient of a curve that is not polynomial; the three mechanisms below are the ways such a coefficient acquires a significance it does not have. The control target (TIC 142874476), on which the chain was developed, was re-run before and after every change reported in this version and reproduced each time; its record and the period-ladder audit are internal quality assurance and are given in Appendix B.

5.1. Formal timing errors

Formal errors against the residuals. The reduced χ^2 of the quadratic fit is the check that the assigned errors describe the residuals. With formal errors it had median 6.35 (25th–75th: 2.08–37.5), was below 2 for only 8 of 31, and left maximum quadratic residuals of 4.4 min (median) to 18 min (75th percentile) against bars of 0.4–0.7 min: not even the more complex model explained the residuals, which locates the problem in the denominator, not the model.

Consequence of the formal-error run. That run — kept in full as `oc_lote_barras_formais/` — returned 24 of 31 targets with $|dP/dt|/\sigma > 3$ and 23 surviving the adversarial test. After the data-derived errors the count was 13 of 31; after the negative-depth guard and the composition correction of Sect. 5.2.2, 10 of 26. The 13-sigma “detection” that headed the formal-error list (TIC 199688472, $+1.75 \pm 0.006 \text{ s yr}^{-1}$) was a quadratic bending to reach two SuperWASP seasons of $\pm 76 \text{ min}$ at 19 yr of leverage, one of which had a negative fitted depth; the target is now in Table 2. The entire positive tail of the earlier distribution (95th percentile $+0.84 \text{ s yr}^{-1}$, now a maximum of $+0.31$ over all 26) consisted of such targets. Formal timing errors in archival photometry produce spurious period changes in bulk, and they produce them preferentially with large leverage, where they look most significant.

5.2. The residual test and its null: what data-derived bars can be validated against

This subsection is about the test that validates data-derived bars — the residuals of the quadratic fits, normalised by the bar of each epoch, corrected for leverage and summed by data set — and its result is a property of the test rather than of the sample. Stated first: when each bar is estimated from the same two or

three samples that enter the residual, the null of the test is 0.77 and not 1 (Sect. 5.2.3); a reduced χ^2 near 0.7 is therefore what correct bars *produce* in this design, and the number to compare with. Measured against that null, the bars of Sect. 3.3 are consistent with the residuals of the 26, with the exception of three targets whose quadratic the goodness-of-fit test rejects, and over the other 23 both archives are inside their nulls once the same cut is applied inside the simulation — a reading the test cannot separate from bars inflated by real secular terms of the tabulated size, which lower the factor in the same way (Sects. 5.2.3 and 5.2.5). The subsection also records how this was reached, because the path passed through an archival excess of 2.73 that three tests treated as real and that was two estimation errors of opposite sign (Sect. 5.2.2); the readings that depended on it are withdrawn in Appendix D.

5.2.1. The data-derived bars against the expected reduced χ^2

With errors from the data (Sect. 3.3) the median reduced χ^2 of the 26 quadratic fits is 0.67. With 1–4 degrees of freedom the median of χ^2/ν under correct bars is well below 1 (0.45 at $\nu = 1$), and for the actual distribution of degrees of freedom of the 26 (five at 1, eight at 2, six at 3, seven at 4) the expected median is 0.73, with a 95% range of the median of 0.43–1.15 (20 000 simulations; `src/reinflar_tess_26.py -resumo`); 0.67 has probability 0.65 of being this high. The aggregate says the same: $\Sigma\chi^2 = 80.1$ for 67 degrees of freedom, reduced χ^2 1.20 ($p = 0.13$). An earlier version measured 1.41 and 1.92 here, read them as bars 1.39 times too small, and built a correction on that reading; the difference is the composition error of the next subsection, and the reading is withdrawn (Appendix D).

5.2.2. The leverage-corrected decomposition by data set, and the composition error it exposed

A global χ^2 says whether the total variance is described, not where a problem sits. The residuals of the quadratic fits, normalised by the bar of each epoch and corrected for leverage — with three parameters on 4–7 points, $E[r^2/\sigma^2] = 1 - h$, and h reaches 0.6 for the TESS epochs — were summed separately over the 84 TESS and 61 SuperWASP epochs of the 26 targets (`src/qual_barra_26.py`; expectation written before running: excess in TESS). The SuperWASP epochs give $\Sigma z^2 = 47.8$ against 35.3 expected under known bars, a variance factor of 1.35 (χ^2 interval 0.89–2.29; bootstrap over targets 0.63–2.38); the TESS epochs give 32.3 against 31.7, a factor of 1.02 (0.66–1.79; 0.42–1.80). The mean leverage is 0.62 for the TESS epochs and 0.43 for the SuperWASP ones: the test retains 38% of the TESS variance and 57% of the SuperWASP variance.

These are not the numbers the first decomposition returned. It returned a SuperWASP factor of 2.73 (1.79–4.64), read as bars 1.65 times too small, and that reading survived a bootstrap over targets, a simulation of the estimator’s leverage bias and an injection of orbits, and was carried through a “corrected” set of detections and a joint correction with the TESS bars. An external reader asked whether 2.73 could be an artefact of the denominator, and the literal check of that question found something simpler underneath: the routine that measures the between-season dispersion returns the standard error of the season *mean*, $\text{std}(d)/\sqrt{n}$, which is the right bar for the single combined epoch used on the control target (Appendix B), and the batch chain, which enters each season as its own epoch, assigned that same

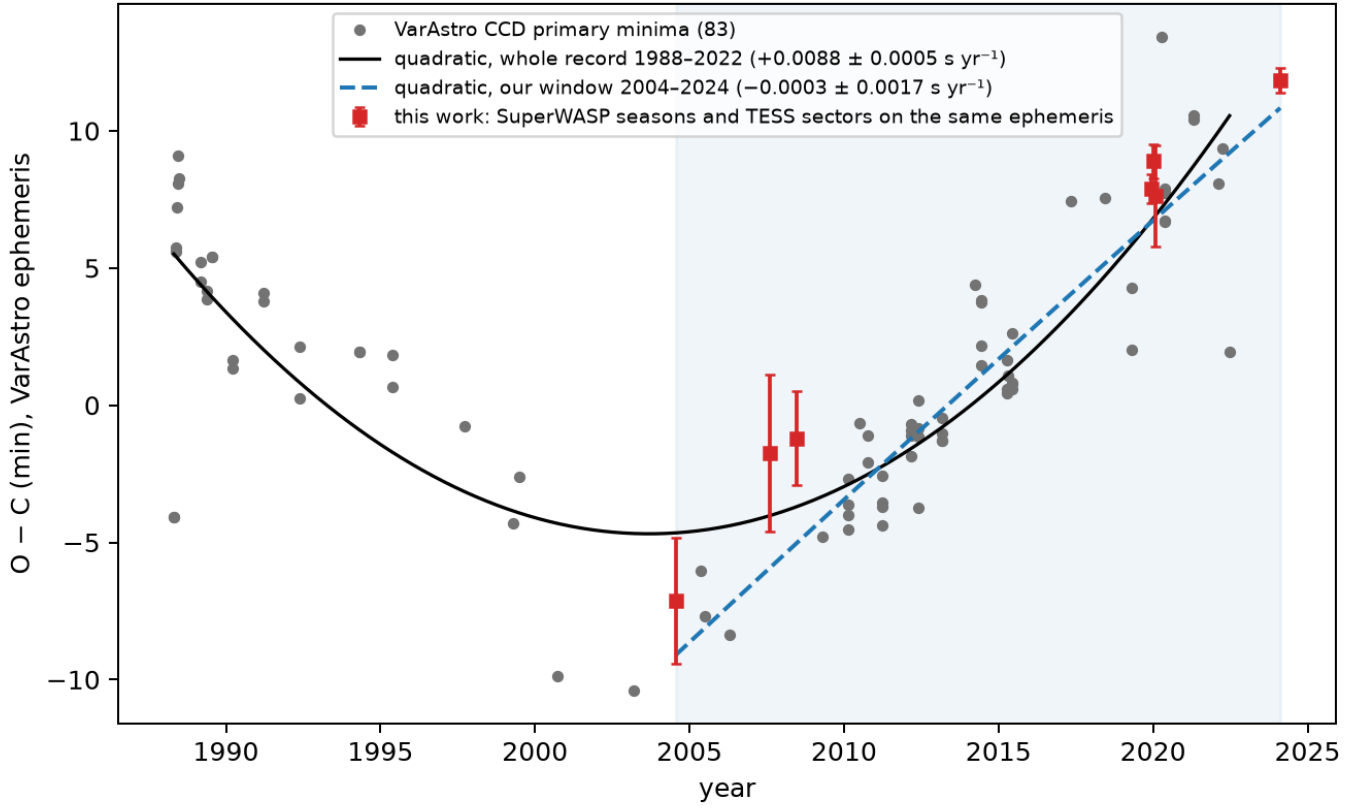


Fig. 2. CV Dra: VarAstro CCD primary minima 1988–2022 with the quadratic fitted to the whole record and the one fitted to our 2004–2024 window, and our epochs placed on the same ephemeris.

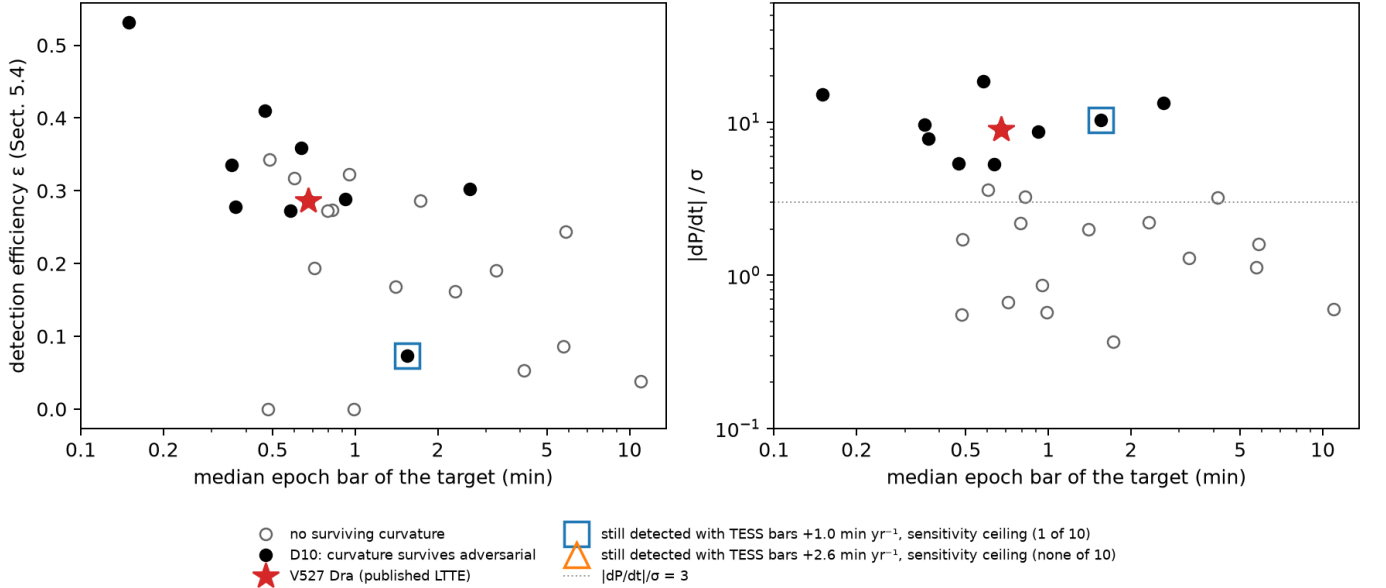


Fig. 3. Detection efficiency, significance and the bars. Left: the per-target detection efficiency ε of Sect. 5.4 against the median epoch bar of the target; right: $|dP/dt|/\sigma$ against the same bar. Filled symbols are D10, the star is V527 Dra; open squares and triangles mark the targets still detected when the TESS bars are inflated by 1.0 and 2.6 min yr⁻¹ (one and none); the dotted line in the right panel is $|dP/dt|/\sigma = 3$.

value to every season. The bar of one season is $\text{std}(d)$. With $n = 2$ seasons in 17 targets and 3 in 9, the assigned bars were $\sqrt{2}$ – $\sqrt{3}$ too small in 49 of the 61 SuperWASP epochs — a factor of 1.74 in variance once weighted as the fit weights them, predicted from the stored records before the code was touched. Corrected (src/oc_lote.py; expectation written and committed before rerunning: the control unchanged, 26 of 88 still measured, the survivors 11 $\rightarrow$ 10 with TIC 359552377 falling, the SuperWASP factor near 1.6), the chain reproduced the control target exactly — the control uses the combined epoch, for which the standard error is correct, which is why it never saw the defect — kept 26 of 88, lost exactly TIC 359552377 ($p_{\text{adv}} 0.05 \rightarrow 0.22$) and no other, moved no coefficient by more than 0.04σ , and brought the median reduced χ^2 from 1.41 to 0.67 and the SuperWASP factor from 2.73 to 1.35: below the expectation of 1.6, because larger bars also change the weights and the parabola re-settles. The between-season dispersion exceeds the formal error by a median factor of 4.3, not 3.0 (Sect. 3.3). The comment above the line said “the dispersion”; the code passed the dispersion over $\sqrt{n}$. It is a defect at the seam between two callers of one function, and the control target, by construction, could not detect it.

The test is asymmetric and that must be said: it has power for SuperWASP, whose epochs sit at the end of the lever, and little for TESS, whose four epochs are fitted by three parameters — what the parabola absorbs does not appear as a residual, and the TESS-internal structure is exactly what was called curvature. A TESS excess at the between-sector scale is therefore not excluded by this test; it is only not required by it. Whether the leverage biases the two factors when the bars are *known* was measured (src/vies_alavanca_26.py; 2 000 realisations per case on the 26 designs): for errors that scale the bars the estimator recovers the truth within 2% in both sets, with no leakage between them; for coherent timing structure at the TESS epochs outside the parabola (a sinusoid of period 2 yr) the TESS factor responds first and strongly (1.25, 3.3 and 10 for amplitudes of 1, 3 and 6 min) and the SuperWASP factor only weakly (1.01, 1.23, 1.89), so the measured TESS factor of 1.02 (≤ 1.79) bounds such structure at about 1 min. What the test cannot see is structure *inside* the parabola — a drift that is linear or quadratic over the four TESS epochs is absorbed into the coefficient — and that is the blindness that matters, because it is exactly the between-sector drift of the secondaries (Sect. 5.3).

5.2.3. The null of the estimator: bars re-estimated from two or three samples

The χ^2 interval quoted above assumes the bars are known. They are not: each SuperWASP bar is the standard deviation of 2–3 season deviations that also enter the residual, and each TESS bar is $\max(\text{formal}, |t_a - t_b|/\sqrt{2})$ from the two halves of the sector — in both cases s is estimated with $\nu = 1-2$ degrees of freedom from the same points as r . The same external reader pointed out that for s independent of r , $z^2 = r^2/s^2$ follows $F(1, \nu)$ with mean $\nu/(\nu - 2)$, undefined for $\nu \leq 2$, and that the bootstrap width was the signature. The null was therefore simulated as the pipeline actually operates (src/nulo_barras_reestimadas_26.py; expectation written and committed before running: a SuperWASP null of 1.3–1.8, the measured 1.35 inside its central interval): on the 26 real designs, Gaussian noise with true σ equal to the assigned bar and no curvature, the bars re-estimated from the simulated seasons and sector halves with the pipeline’s own rule, and the leverage-corrected sums formed with those bars; 2 000 realisations.

The expectation was wrong in direction. With bars re-estimated, the null of the SuperWASP factor is 0.77 (2.5–97.5%: 0.63–0.93; quartiles 0.72–0.81), not above 1, and the TESS null is 0.96 (0.44–1.91). The $\nu/(\nu - 2)$ argument assumes r independent of s ; here they are the same numbers. With two seasons the parabola passes through their mean, the residual of each is $\pm(d_1 - d_2)/2$ and the bar is $|d_1 - d_2|/\sqrt{2}$, so $z^2 = 0.5$ by construction, and the floor (the formal error, the 0.78 min) pushes it lower still. With the bars held fixed the same simulation returns 1.00 in both sets — the estimator is unbiased when σ is known and biased low when σ is co-estimated, and the second case is the pipeline. Treating the 0.78 min as a common offset of the SuperWASP block rather than as per-season noise changes nothing (0.76). A reduced χ^2 near 0.7 is therefore what correct bars *produce* in this design; it is the number to compare with, not a sign of over-estimated bars. The values themselves — 0.77, and the 0.70 of the conditioned TESS null below — are properties of this design (26 targets, 4–7 epochs each, bars co-estimated with $\nu = 1-2$ and floored by the formal error and the 0.78 min) and not constants: another sample, another number of seasons per target or another floor gives another null, and the simulation is what transfers, not the value.

Against its own null the measured SuperWASP factor of 1.35 is not a small- ν artefact but an excess: none of the 2 000 realisations reaches it. Where it sits is a reading made after the fact and labelled as such, but the criterion it uses is the one Sect. 3.5 declares, not one chosen after seeing the residuals: the three targets whose quadratic the goodness-of-fit test rejects ($p_{\text{gof}} < 0.05$; TIC 198388252, 229914020, 237116051) contribute $\Sigma z^2 = 13.3$ (237116051, two epochs) and 11.5 (198388252, three) of the 47.8. Over the 23 targets with an adequate quadratic the SuperWASP factor is 0.73 and the TESS factor 0.55. The null those numbers are compared with must be conditioned in the same way, because the 23 are the targets that passed the goodness-of-fit cut: the cut was therefore applied inside each realisation (p_{gof} of every target’s quadratic computed with the re-estimated bars, the targets below 0.05 removed, the factors formed over the survivors; an earlier version compared with the null of the 23 designs without the cut, 0.78, which is not the same experiment — the external reader pointed this out). Under the null the cut removes 0.83 targets per realisation — 3.2% per target rather than 5%, because the re-estimated bars deflate χ^2 — and three or more in 5.7% of realisations, so the count of flagged targets is itself a mild excess. The conditioned nulls are 0.77 (0.63–0.92) for SuperWASP, unchanged, and 0.70 (0.37–1.08) for TESS, well below the unconditioned 0.96: the realisations that fail the cut are those in which a TESS bar co-estimated from two half-sectors came out small, and they hold 31% of the TESS Σz^2 of the null in 3% of the target-realizations (expectation before running: a TESS null of 0.85–0.93; the size of the drop was not anticipated). The measured 0.73 is at the 30th percentile of the SuperWASP null and the measured 0.55 at the 23rd of the TESS null. The excess is not in the archival bars; it is in three targets whose parabola does not describe the data, and one of them, TIC 198388252, is a member of D10 — a detection that passes the significance and adversarial tests on a quadratic the goodness-of-fit test rejects, the same structure as V527 Dra with a different confounder (Sect. 5.4). No bar correction is applied: the bars of Sect. 3.3, with the composition error corrected, are the bars of Tables 3–5, and the earlier “corrected” and “jointly corrected” sets of detections are withdrawn (Appendix D).

What the test can and cannot separate must then be said. A real secular term inflates the co-estimated archival bar

(Sect. 5.2.5), and an inflated bar lowers the factor: with a term injected in every target (`src/completude_secular_26.py` -factor; expectation written and committed before running: a monotonic fall, about 0.70 at 0.02 s yr^{-1}) the SuperWASP factor with the goodness-of-fit cut inside is 0.77, 0.72, 0.60, 0.44 and 0.30 at 0.005, 0.01, 0.02, 0.05 and 0.2 s yr^{-1} , while the TESS factor stays at 0.69–0.73, because half-sector bars do not see the term and the parabola absorbs it exactly (with the bars held fixed the factor is 1.00 at every amplitude: the term lies in the model space and the whole effect is the re-estimation). The measured 0.73 of the 23 sits at the 33rd percentile of the null and at the 56th of the distribution with 0.01 s yr^{-1} in every target, and $P(\leq 0.73)$ is 0.87 at 0.02 s yr^{-1} and 1.00 at 0.05: “the bars describe the residuals” and “the targets carry real terms of up to 0.02 s yr^{-1} that inflated their own bars” give the same reading, and the test does not separate them; what it does exclude is a term of 0.05 s yr^{-1} in every target, which would have put the factor below 0.73 in every realisation. Nor is the excess of the 26 immune. Without the cut the *mean* of the factor rises — 0.80, 0.84 and 0.83 at 0.005, 0.01 and 0.02 s yr^{-1} — while its median does not (0.78, 0.79, 0.73), so the rise is a tail: 2.5% of realisations above 1.5–1.9, against 0.92 with no term. The reading first written for it, that the inflated archival bars hand the fit to the TESS block, was tested and is wrong (the mean leverage of the archival epochs stays at 0.42–0.43 across the grid); what the term breaks is the coupling that makes the null 0.77. With two seasons the bar is $|\Delta|/\sqrt{2}$ and the residual $\pm\Delta/2$ of the same between-season difference Δ , so $z^2 = 0.5$ by construction; with a term, the bar is built from the signal’s between-season difference plus the noise’s, and the residual — the parabola having absorbed the term — from the noise’s alone, so that when the two nearly cancel the bar drops to its floor while the residual does not. The probability of a factor ≥ 1.35 is 0.001 with no term and 0.04–0.08 with 0.01 – 0.05 s yr^{-1} in every target: a population of real terms of the tabulated size is not excluded as the source of the excess of the 26, although the three flagged targets remain the simpler account of where it sits. What this subsection can say is therefore narrower than “the bars are right”: the bars are consistent with the residuals, and so would be bars inflated by real terms of up to about 0.02 s yr^{-1} in every target.

5.2.4. What the residuals leave for orbits, with the same estimator

An undersampled third-body orbit is the structure that the four TESS epochs absorb into the parabola and the SuperWASP seasons, 12–16 yr away, do not, and V527 Dra shows the sample contains at least one. How much of an archival excess such orbits could produce was measured by injecting the declared population of Sect. 5.4 into the 26 designs with occurrence f and reading the two factors (`src/estructura_swasp_26.py`). Done with the bars held fixed, as first run, the injection answers a question the measurement does not ask — its null is 1.00 and the measurement’s is 0.77 — and the numbers it gave (orbits at $f = 0.24$ explaining 42% of the excess; a bar-only factor of 1.54; a TESS-side bound of $f \leq 0.5$) are withdrawn as cross-estimator comparisons (Appendix D). Re-run with the bars re-estimated inside the simulation (expectation written and committed before running: the null reproduced at $f = 0$; a SuperWASP factor of 0.95–1.10 and a TESS factor near 1.5 at $f = 0.24$), the injection reproduces the null at $f = 0$ (0.77) and gives, at $f = 0.24$, a SuperWASP factor of 0.84 (0.65–1.19) and a TESS factor of 1.42 (median 1.23): the re-estimated bars absorb most of an orbit’s

between-season signal, so the archival side is nearly blind to orbits — below the expectation of 0.95–1.10 — and the TESS side is where they show. The probability that orbits alone produce the 26-target SuperWASP factor of 1.35 is 0.002 at $f = 0$, 0.021 at $f = 0.24$ and 0.16 at $f = 1$: the excess of the three flagged targets is not an orbit population. Over the 23 adequate targets the comparison is conditioned as in Sect. 5.2.3 — the goodness-of-fit cut applied inside each realisation, with the orbits injected, and the factors read over the survivors (at $f = 0$ this reproduces the conditioned nulls of Sect. 5.2.3, 0.76 and 0.72 against 0.77 and 0.70 there: independent Monte Carlo runs of the same quantity) — and the SuperWASP side constrains nothing ($P(\geq 0.73 | f) = 0.68$ – 0.92). The TESS side of the 23 is nearly blind as well, for a reason worth stating: the cut removes the targets whose orbit makes the parabola inadequate, 0.8 per realisation at $f = 0$ and 4.0 at $f = 1$, so the surviving set loses the orbits it is being asked to detect; $P(\text{TESS} \leq 0.55 | f) = 0.20, 0.19, 0.13, 0.10, 0.068$ and 0.027 at $f = 0, 0.10, 0.24, 0.41, 0.70$ and 1.0, which excludes only $f \approx 1$. The TESS side, with the estimator now matched, bounds the in-window fraction from above: $P(\text{TESS} \leq 1.02 | f) = 0.64, 0.49, 0.34, 0.20, 0.073$ and 0.017 at $f = 0, 0.10, 0.24, 0.41, 0.70$ and 1.0, so the TESS side excludes only $f \geq 0.8$ at 95% — above every published in-window fraction, i.e. an uninformative bound — and 0.24 and 0.41 are both compatible. The test is one-sided — p decreases monotonically with f and cannot identify a central value — and the central value is taken from Tokovinin et al. (Sect. 5.4), not from it.

5.2.5. A secular term through the same estimator: completeness and self-suppression

The two subsections above ask what the estimator does to noise and to orbits; this one asks what it does to the signal the paper is nominally about. A pure secular term, $t(E) = QE^2$ in the convention of Table 3, was injected on the 26 real designs at $|dP/dt| = 0.005$ – 0.25 s yr^{-1} with random sign (the median tabulated $|dP/dt|$ is 0.016), with Gaussian noise of the assigned bars, the bars re-estimated inside each realisation by the pipeline’s rule — for SuperWASP the standard deviation of the simulated seasons *with the signal in them*, because the pipeline measures each season against the TESS period and a real secular term appears as season-to-season spread — and each realisation judged by the curvature test and the adversarial gate (`src/completude_secular_26.py`; 300 realisations per target and amplitude; two expectations written and committed before running, the external reader’s and the author’s, which differed on the size of the suppression). Fig. 4 shows the two curves of every target.

Completeness. With no term injected the gate fires in 0.2% of realisations at the median target, in at most 6% in 24 of the 26, and in 16% for TIC 229476285, whose 359-min archival bars leave the fit to its three TESS epochs; summed over the 26 the false-alarm rates give 0.36 detections expected from noise alone, and no member of D10 has a rate above 1%. The amplitude at which half the injected terms are detected has a median of 0.015 s yr^{-1} — the median tabulated coefficient — and completeness at 0.02 s yr^{-1} is ≥ 0.5 in 16 of the 26; at 0.2 s yr^{-1} it exceeds 0.9 in 21 and stays below 0.75 in the five targets with archival bars above 8 min (TIC 126763885, 139256217, 224605072, 229476285, 359552377), three of which never reach 0.5 on the grid. One of the three, TIC 359552377, is the target that left the detected set when the composition error was corrected: at the coefficient it carries, -0.25 s yr^{-1} , the gate passes 26% of injected terms on its design (17% with the bars held

fixed; the suppression factor there is 0.95 — its 16-min bars are not inflated by the term, they are simply large at a 15-yr lever), so its departure is as compatible with a real term the chain cannot hold as with a spurious one it let go (Appendix D). Over the 23 that reach it, the 50% amplitude correlates with the median archival bar (Spearman +0.42, $p = 0.05$) and, more weakly and in the opposite sense, with the number of epochs (-0.32 , $p = 0.13$); with 23 targets, and bar size and epoch count themselves correlated, the two are not separated here, and the three targets that never reach 0.5 are the three with the largest archival bars. At the coefficient each of them carries in Table 3, the ten members of D10 have a completeness of 0.54–1.00 (median 0.88): the denominator of “ten detections” is not ten, and no count of detections in this paper is a rate. Against the same tests with the bars held fixed, re-estimating them costs 25 points of completeness at 0.02 s yr^{-1} (median 0.65 against 0.90) and nothing from 0.05 upward.

Self-suppression. The re-estimated bars contain the signal, and the signal inflates them. Against the expectation of a 20–40% effect at 0.2 s yr^{-1} , $\sigma(dP/dt)$ with re-estimated bars exceeds σ with the true bars by a median factor of 1.15 at 0.02 s yr^{-1} , 1.26 at 0.05, 1.6 at 0.1 and 2.8 at 0.2 — and by 5–24 at 0.2 in the nine targets whose archival seasons have bars of 1.5–5 min at the longest lever (TIC 198408416 $\times 24$, 198388252 $\times 16$, 229687624 $\times 13$, 377253090 $\times 12$, 329246824 $\times 11$, 390021728 $\times 9.5$, 329289186 $\times 8.3$, 144194304 $\times 8.2$, 199688409 $\times 5.0$). The simulated noise is the assigned bar of each epoch, which on a target with real structure already contains it, so the simulated noise is there larger than the true one and the suppression measured is a floor. The reason for it is arithmetic: a term of 0.02 s yr^{-1} at $E \approx -8000$ cycles moves two seasons 730 cycles apart by about 3 min relative to each other, the size of the bars; at 0.2 s yr^{-1} the between-season spread of the signal is tens of minutes and the “bar” is made of it. The coefficient itself is unbiased (median recovered/injected 1.00 at every amplitude) and the goodness-of-fit cut does not see the term ($p_{\text{gof}} < 0.05$ in 2–4% of realisations at the median, at most 21% in one four-epoch design): the estimator cannot manufacture a detection from a secular term, but it turns the tabulated σ of a real one into an overestimate by a factor that grows with the term, and it hides real terms near the 50% point. At the coefficients of Table 3 the factor for the ten members of D10 is 1.03–1.73 (median 1.4). This is an estimator property of the same family as the composition error of Sect. 5.2.2 — a consequence of estimating each season’s bar from the seasons themselves — and, unlike it, not correctable by a rule: the alternative, bars from the formal errors, is the one Sect. 5.1 shows reporting 24 significant targets of 31. Two readings follow. The nine detections carried by TESS are not affected, because the TESS bars come from half-sectors, which no secular term can spread. For TIC 232634196 the simulation on its own design, with the noise taken as its 19.8-min bars, gives a completeness of 0.94 and a suppression of 1.7 at its coefficient; if those bars were instead the signal’s own spread on the 1.4-min formal noise — which is what the simulation produces for a term of that size on the small-bar designs — the tabulated σ would be an overestimate by an order of magnitude. The data decide against the second reading on one point: if the 19.8 min were the term’s own spread on 1.4-min formal noise, the three seasons would fall on the fitted parabola within about 1.4 min, and they fall at 13, 0 and -22 min from it; the prediction of Sect. 5.5 is written with the tabulated σ .

5.3. Between-sector timing noise: the secondaries and the TESS sensitivity ceiling

5.3.1. Secondary minima: the differential drift between sectors

Apsidal motion in an eccentric orbit produces a spurious quadratic in the primary alone, with the secondary moving the opposite way; a secular dP/dt moves both together. The secondary epoch was measured in every TESS sector of the 26 with the same estimator, the catalogue seed shifted by $P/2$, the same trapezoid and the same halves-dispersion bar (src/secundarios_26.py; expectation written before running: most secondaries measurable, 0–2 targets with a significant differential drift, up to 5 with no measurable secondary). The quantity tested is $d = t_{\text{sec}} - t_{\text{prim}} - P/2$ per sector, and its drift between the first and last sector (median lever 4.2 yr within TESS).

All 26 secondaries were measurable; for two of the long-period targets (TIC 139256217, 359552377) the bar on d exceeds 10 min and the test is uninformative, leaving 24. In 5 of the 24 the secondary sits more than 2σ from phase 0.5 (TIC 126763885, 224605072, 229914020, 232634196, 329246824), which is a statement about eccentricity or asymmetry, not about period change. In 5 of the 24 the differential drift is significant at $> 2\sigma$ — more than the 0–2 expected: TIC 229476285 ($-1.50 \pm 0.38 \text{ min yr}^{-1}$), 229914020 ($+2.59 \pm 0.29$; flagged), 232634196 (-1.80 ± 0.50), 329289186 ($+2.41 \pm 0.52$) and 390021728 ($+0.67 \pm 0.33$). None of the four targets with $P > 2$ d and an informative secondary shows a drift, so the eccentric-orbit confounder is not detected where it was expected; the drifts occur in systems with $P = 0.3$ –1.25 d, where migrating spots are the usual cause of differential primary/secondary timing.

On the one target where a published primary/secondary comparison exists, HD 202042 (TIC 126945917), Pawar et al. (2026) report an anti-correlation within TESS sectors; our sector-to-sector drift for the same target is $+0.50 \pm 0.49 \text{ min yr}^{-1}$, not significant, so our data do not confirm their effect at the level our bars allow.

Of the ten detections of D10, two are affected, and they are one story. TIC 229914020 has the largest differential drift of the sample ($+2.59 \pm 0.29 \text{ min yr}^{-1}$, 9σ), is flagged by the goodness-of-fit test, and is in D10: its curvature, its bad fit and its drifting secondary are three views of the same non-quadratic timing behaviour, and the target belongs to none of the categories that read as period change. TIC 232634196, in D8, has a differential drift of $-1.8 \pm 0.5 \text{ min yr}^{-1}$ measured over 4.2 yr, i.e. 7.5 min within the TESS span; extrapolated linearly over the 20-yr baseline — an illustration, since the drift was measured on a fifth of that interval — it would amount to 36 min against a quadratic excursion of 208 min, so the secondary shares most of that curvature, not necessarily all of it. The primary/secondary test therefore qualifies two of the ten; it does not bear on the third-body mechanism of Sect. 5.4, which moves both minima together.

The excess itself — five drifts against 0–2 expected — is the measurement behind the TESS sensitivity test below: a spot-driven drift of 1 – 2.6 min yr^{-1} between sectors years apart is invisible to a within-sector bar and to the residual test, and when a drift of that size is added to the TESS epochs, nine of the ten detections disappear. Whether the primaries drift as the secondaries do is the question the data leave open.

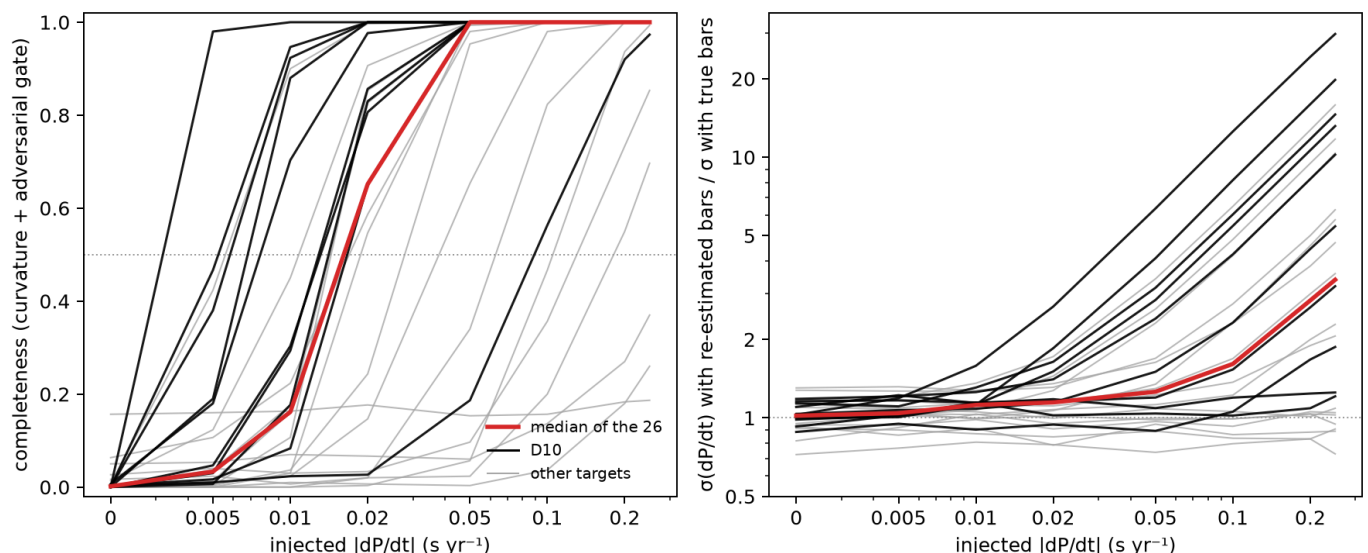


Fig. 4. A secular term through the estimator of the chain (Sect. 5.2.5). Left: completeness — the fraction of injected terms passing the curvature test and the adversarial gate with the bars re-estimated inside the simulation — against the injected $|dP/dt|$, one curve per target (black: D10; grey: the others; red: the median of the 26; the point at 0 is the false-alarm rate). Right: the suppression factor, $\sigma(dP/dt)$ with the re-estimated bars over σ with the true bars, on the same grid.

5.3.2. TESS bars at the between-sector scale: a sensitivity test with a ceiling

The secondary-minimum test above finds the secondary drifting with respect to the primary by $1.0\text{--}2.6\text{ min yr}^{-1}$ between sectors in five of 24 targets. The TESS bar of Sect. 3.3 is the dispersion between the two halves of a sector, a within-sector quantity over ~ 13 d, and cannot contain such a drift; the residual test above cannot see it either. Whether the primary epochs wander at a comparable rate is therefore untested, and the question matters because the detections are TESS-internal.

The test that was run (`src/reinflar_tess_26.py`; expectation written and committed before running) inflates every TESS epoch bar of a target in quadrature by $d \times L$, with $d = 1.0$ or 2.6 min yr^{-1} (the extremes of the measured secondary drift; no central value) and L the target’s own TESS lever (median 4.2 yr), giving median TESS bars of 4.2 and 10.9 min; the SuperWASP bars are unchanged; the stored epochs are refitted (linear and quadratic, curvature and adversarial tests, goodness of fit, the TESS-only period and the cycle-count criterion) and the efficiency of Sect. 5.4 is re-measured. This is an extrapolation of a drift measured on the secondary in five targets to the primary epochs of all 26, and 1.0 min yr^{-1} is a ceiling, not an estimate: the TESS residuals do not ask for it, and an earlier version of this paper, which called the 1.0 min yr^{-1} bars “calibrated” because they brought the median reduced χ^2 to 0.73, was wrong — that coincidence came from re-weighting (larger TESS bars let the parabola reach the SuperWASP seasons, reducing their residuals), not from correcting the bar the residuals identify. The expectation, written for the eleven detections of the earlier bars, was that 6–9 would survive at 1.0 min yr^{-1} and 3–6 at 2.6. The result, re-run on the corrected bars: **at 1.0 min yr^{-1} , one of the ten survives (TIC 232634196); at 2.6, none** (Table 5; Table 4; Fig. 3). The cycle count across the gap stays unambiguous in every case ($\sigma_P \times N \leq 0.04$).

The response is not monotonic, and “no new detection appears” would misdescribe it: at 1.0 min yr^{-1} four targets have $p_{\text{curv}} < 0.05$ (TIC 232634196, 237116051, 359552377,

392536812) and only the first passes the adversarial test; two targets become *more* significant in curvature as the bars grow — TIC 237116051 from $p_{\text{curv}} 0.39$ to 0.001 and TIC 198388252 from 0.11 (at 1.0) to 0.002 (at 2.6) — because the weight migrates to the SuperWASP seasons and the fit changes, and both fail the adversarial test. A reading, generated after the fact and labelled as such: inflating the TESS bars does not only remove detections, it redistributes the weight between the two time-scales, and for those two targets a long-baseline curvature that the TESS weight suppressed emerges when the archival seasons dominate. The reading is not delivered by the test — both targets fail the adversarial test, and the same redistribution story would fit any target whose weights migrate — so it is a hypothesis about where each target’s curvature lives, not a result.

Nine of the ten detections are therefore structure within the five years of TESS, and the paper cannot certify that structure as period change against between-sector timing noise of the size the secondaries show: the within-sector bars do not contain it, and the residual test cannot detect it.

5.4. Undersampled third-body orbits

V527 Dra: a false positive of secular curvature that passes every internal test. The eclipse-timing analysis of V527 Dra (Čeki et al. 2024; 1 832 timings including 34 from SuperWASP) is a light-travel-time orbit with $P_3 = 2.733 \pm 0.005\text{ yr}$ and semi-amplitude 6.6 min, *with no secular term*; the star is also one of the eleven objects SIMBAD links to the TESS full-frame-image timing study of Marcadon & Prša (2024), which reports ten new triple candidates. Our chain, with six epochs and no access to any of those timings, returns $dP/dt = -0.0359 \pm 0.0040\text{ s yr}^{-1}$ with $p_{\text{curv}} \approx 10^{-19}$, $p_{\text{adv}} \approx 10^{-12}$, $p_{\text{gof}} = 0.11$: a curvature detection that survives the adversarial test and whose quadratic fit is not rejected, in a system whose published O–C has no secular term. **The one target with an external record of a cyclic O–C passed all three internal tests, and it is a member of D8, the cleanest set this paper has.** Only two members of D8 have an external record long enough to test the secular reading, V527

Dra and V564 Dra (Sect. 5.5); one fails and the other agrees only within our window, and the other six are untested, not confirmed. V527 Dra is an existence proof; the rate comes from injection, below. Two earlier readings of this target — a “blind validation” of a fixed $\chi^2_{\text{red}} \geq 2$ flag, and a false-positive rate of “one in nine verified” — are withdrawn (Appendix D).

The mechanism is measurable. Six epochs over 15.7 yr sample a 2.73-yr cycle at essentially three phases (2008.4–2008.6, 2019.9–2020.1, 2024.1), and a three-parameter parabola passes through three groups, leaving only the intra-group structure as residual: this is why the goodness-of-fit test does not reject the parabola ($\chi^2_{\text{red}} = 2.03$ against an expected median of 0.79 at 3 degrees of freedom — within the tail), whereas a 6.6-min signal against sub-minute bars would give χ^2_{red} of tens if the cycle were not absorbed. The published orbit (circular, $A = 6.6$ min, $P_3 = 2.733$ yr) was injected on an exactly linear ephemeris at our six epoch times with our bars, over 360 orbital phases (src/v527_ltte.py; expectation written before running; Fig. 5). The quadratic coefficient ranges from -0.058 to $+0.058$ s yr $^{-1}$ ($|dP/dt| \geq 0.036$ at 57% of phases), $p_{\text{curv}} < 0.05$ at 91% of phases, $p_{\text{adv}} < 0.05$ at 82%, χ^2_{red} median 2.83 (below 3 at 52% of phases, maximum 5.6); 25 of the 360 phases return a coefficient within 0.005 s yr $^{-1}$ of the observed one, with χ^2_{red} between 1.3 and 5.2. The observed outcome is a typical product of a parabola over an undersampled third-body cycle. **The adversarial test shifts epochs, not phases, and does not protect against it: 82% of phases survive.** Any triple system in the sample with an outer period shorter than the baseline and longer than a sector passes every internal test here and fails only an external one. The rest of this subsection asks how many such systems the sample is expected to contain.

The V527 Dra result turns a case into a rate question, with two ingredients that must not be confused: how often a close binary has a tertiary with an outer period inside the window where undersampling produces curvature, and how often such an orbit, sampled at a target’s actual epochs with its actual bars, produces a detection by the tests of Sect. 3.5.

Detection efficiency per target. The 82% measured above is for one orbit and one sampling pattern, and cannot be transferred. The efficiency was measured on each target (src/epsilon_26.py; expectation written and committed before running) by injecting a declared population of circular tertiary orbits — P_3 log-uniform in 0.1–20 yr, masses and inclination as in Appendix C, amplitude from Kepler’s third law, which reproduces the published 6.6 min of V527 Dra — on an exactly linear ephemeris at the real epoch times with the real bars, 2000 orbits per target, each judged by the curvature and adversarial tests. ε ranges from 0.00 to 0.53 (median 0.27, mean 0.23), is set by the size of the bars and not by the sampling pattern (Appendix C), and sums to $\Sigma\varepsilon = 6.1$ over the 26 with the bars of Sect. 3.3 (6.7 with the bars before the composition correction; 21.6 is the ceiling for orbits all as favourable as V527 Dra’s). It treats orbit and secular term as exclusive, and it is measured with the bars held fixed: an orbit that spreads a target’s seasons also inflates the bars the pipeline would co-estimate from them (Sects. 5.2.4 and 5.2.5), so the pipeline’s own efficiency is at most this one — the archival-lever part of ε shrinks under re-estimation, the TESS-internal part, whose bars come from half-sectors, does not — and $\Sigma\varepsilon$ and every number built on it below are ceilings. Two targets (TIC 224605072, 229476285) have $\varepsilon = 0$: their SuperWASP bars are too large for the curvature test to reject anything.

In-window tertiary fraction. From Table 5 of Tokovinin et al. (2006; VizieR J/A+A/450/681), of the 42 solar-type spectroscopic binaries with inner period below 3 d, 10 have their

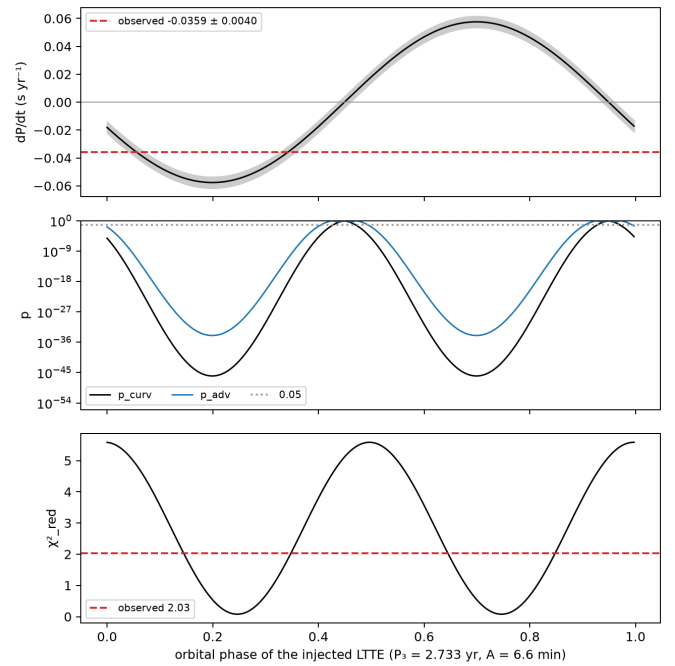


Fig. 5. The published LTTE orbit of V527 Dra injected on a linear ephemeris at our six epochs, over 360 orbital phases: the resulting quadratic coefficient (top), the curvature and adversarial p-values (middle) and the reduced χ^2 (bottom) against phase, with the observed values marked.

closest tertiary at an outer period of 0.1–20 yr: $f = 0.24$, 95% Wilson interval 0.13–0.39 (0.41 under the hypothesis that all the incompleteness the paper corrects for lies inside the window). The TESS residuals of this sample give an independent bound, with the estimator matched to the measurement (Sect. 5.2.4): the declared population is compatible with $f = 0.24$ (probability 0.34 of a TESS factor as low as measured) and with 0.41 (0.20), and $f \geq 0.8$ is excluded at 95%. The internal test is one-sided and bounds f from above only, which is why the central value is taken from Tokovinin et al. and not from it. Survey-detected fractions (Borkovits et al. 2016; Hajdu et al. 2019; Marcadon & Prša 2024) are floors already convolved with each survey’s efficiency and are not multiplied by ε (Appendix C). The transfer to this sample — six targets with $P > 2$ d, five detached — is asserted, not demonstrated.

The number. The expected number of the ten detections produced by undersampled orbits is $f \times \Sigma\varepsilon = 0.24 \times 6.1 = 1.5$ over the 26 (0.8–2.4 over the interval on f ; at most 4.9 at the internal ceiling $f = 0.8$), and one of the ten is such an orbit (V527 Dra). **Orbits account for at most one or two of the ten.** The residual, 8–9 detections (± 1.2 , Poisson on the subtracted number), is TESS-internal structure that this paper cannot certify as period change, spot-driven drift or between-sector noise (Sect. 5.3), and one of them (TIC 198388252) sits on a quadratic the goodness-of-fit test rejects (Sect. 5.2.3). For TIC 232634196 a third body is not excluded: $f \times \varepsilon$ is 0.02 under the declared population and 0.2 for a V527 Dra-like orbit.

5.5. External records: published work, archival minima, and the one detection carried by the lever

Published work on the 26. Each target was cross-matched by position with SIMBAD (5″; 26 of 26 matched, median 10 references per object, minimum 1, maximum 33) and with the VarAs-

tro star catalogue (1'; Paschke & Brát 2006), the successor of the O–C Gateway that inherited its minima database (13 of 26 have an entry; 12 have at least one primary minimum; 6 have ≥ 4 CCD primary minima, spanning 7.7–34 yr). The SIMBAD bibliographies are catalogue-scale except for five object-scale papers (≤ 10 objects): TIC 424461577 = V527 Dra (see below); TIC 144194304 = V Gru (Tanriver et al. 2023), whose timing analysis covers 2018–2020 only and reports a linear ephemeris; TIC 229687624 = CV Dra (IBVS 3213, 1988), light curves and preliminary elements; TIC 230386284 (Pyatnytsky 2020), photometry only; and TIC 126945917 = HD 202042, one of five oscillating Algols in Pawar et al. (2026), whose O–C analysis uses TESS sectors 1, 27, 28 and 68 only, attributes an anti-correlation between primary and secondary timings to spots, and states that with data spanning so short a time no conclusion on the mass-transfer rate can be drawn — no dP/dt is reported (ours: $+0.0084 \pm 0.0035$ s yr $^{-1}$, no significant curvature). No target has a published dP/dt . The subpopulation is more studied than we anticipated: the expectation written before the query was ≤ 5 targets in the VarAstro database and ≤ 2 with an object-scale paper.

Independent minima. For the six targets with ≥ 4 CCD primary minima in the VarAstro database, the same quadratic ephemeris was fitted to those minima alone (VarAstro’s own ephemeris for the cycle count; unweighted least squares; per-point error from the MAD of the quadratic residuals; primary minima only, CCD/photoelectric only). One was excluded for a period-convention discrepancy that anyone repeating the cross-match will meet: for V391 Dra (TIC 237116051) VarAstro adopts half the catalogue period (1.2315 d against 2.4629 d), and 5 of its 7 “primary” minima fall at phase 0.5 of the catalogue ephemeris — they are secondary minima under the convention of Table 3, and the two sets do not time the same event. Over the whole VarAstro record, 2 of the remaining 5 agree with Table 3 within 2σ (BL Dra = TIC 229461186, $z = +1.0$; V353 Dra = TIC 207496824, $z = -1.0$) and 3 do not: CV Dra (TIC 229687624; record 1988–2022, 83 minima: $+0.0088 \pm 0.0005$ s yr $^{-1}$ against our -0.0006 ± 0.0021 , $z = -4.3$), V564 Dra (TIC 329246824; 1999–2025, 10 minima: -0.0066 ± 0.0011 against -0.0199 ± 0.0018 , $z = -6.4$) and V527 Dra (the published LTTE system, $z = -6.5$). Restricting the VarAstro fit to the baseline of our own epochs (± 0.5 yr) changes the picture: CV Dra -0.0003 ± 0.0017 ($z = -0.1$), V564 Dra -0.0130 ± 0.0042 ($z = -1.5$), BL Dra $+0.0082 \pm 0.0098$ ($z = +0.1$), V353 Dra -0.02 ± 0.13 ($z = +0.3$) — 4 of 4 (V527 Dra has 2 in-window minima). At the level of the data, placing our epochs on VarAstro’s ephemeris next to the CCD minima taken within one year gives, for the SuperWASP era (the independent one; 2 comparisons, both CV Dra): -7.1 ± 2.3 vs -6.9 min and -1.2 ± 1.2 vs -4.8 ± 1.9 min; for the TESS era, 12 of 13 within 2σ with rms $z = 1.16$ (VarAstro minima after 2018 may themselves derive from TESS, so this is consistency, not independence). The median offset, $+0.5$ min, is of the order of the HJD_UTC/BJD_TDB conversion and is not interpreted.

The window-restricted test is the demonstration: it is not that our measurement disagrees with the archive’s, it is that the two fits measure different windows of a curve that is not polynomial — same window, 4 of 4 agree; whole record, 2 of 5 (Fig. 2). The epochs agree and the coefficients do not, because the O–C is not a parabola over the longer record: the VarAstro O–C of CV Dra runs from $+5$ min (1988) through -10 min (2002) to $+8$ min (2022), and its quadratic residuals reach 9.6 min against an intra-year scatter of 2.4 min. A quadratic fitted over 16–20 yr returns the local coefficient over that baseline, and the two targets with longer external records show that it does not extrapolate.

TIC 232634196: the one detection carried by the archival lever. TYC 3906-474-1 (Tmag 9.9, $P = 1.2511$ d, catalogue morphology 0.66), coefficient -0.214 ± 0.021 s yr $^{-1}$. Unlike the other nine, its curvature is carried by the 20-yr lever: on the 20-yr linear ephemeris of the primaries the three SuperWASP seasons sit at -134 , -98 and -107 min, and the quadratic through them predicts the TESS-internal change that is observed (quadratic residuals of the four TESS epochs within ± 0.7 min). The spread of the three seasons, 36 min, is not a test of them: the 19.8-min bars are the standard deviation of that same spread. What sustains the lever is the *mean* of the three seasons, about -110 min from the linear ephemeris against bars of 20 min, and that is why the composition correction of Sect. 5.2.2, which raised those bars from 11.5 to 19.8 min, moved the coefficient from -0.213 ± 0.013 to -0.214 ± 0.021 s yr $^{-1}$.

It is the only detection that does not depend on the TESS-internal structure: it survives the TESS sensitivity test at 1.0 min yr $^{-1}$ (-0.200 ± 0.031) and fails only at 2.6 (adversarial), and it is insensitive to the archival bars within any factor the residuals allow (p_{curv} and $p_{\text{adv}} < 10^{-4}$ with the bars multiplied by 1.5). The cycle count across the gap is unambiguous ($\sigma_P \times N = 0.014$), there is no VarAstro record and no literature on the star.

Whether the primary and the secondary move together or in opposition was measured on the same 20-yr linear ephemeris of the primaries (src/caso_232634196.py; the expectation was declared as not blind, because the stored numbers were already in view): the change from the 2019–2020 sectors to sector 75 is -2.7 ± 0.9 min for the primary and -11.6 ± 1.5 min for the secondary — same sign, both significant (the values depend on the reference ephemeris, which moved with the corrected bars; the differential does not). That excludes the anti-correlated signature of apsidal motion, and it is where the conclusion stops: the common mode, -2.7 min, is the *smaller* of the two terms, and the differential, -9.0 ± 1.7 min, the larger. A period change plus a differential drift reproduces the observation; so does a spot pattern that affects the two minima unequally. The data do not separate the two hypotheses. The reference matters and is stated: a linear ephemeris fitted to the TESS epochs of both minima at once would split the differential drift between them by construction and read “opposition” for any target; the reference used is the ephemeris against which the curvature was claimed.

What the paper can say about this star is therefore: a curvature over 16–20 yr carried by the archival lever, insensitive to the archival bars and surviving the TESS sensitivity test at its lower value, whose secondary moves in the same direction as its primary, and whose common-mode change is smaller than the differential one; a third-body orbit is not excluded (Sect. 5.4). It is one object, and instead of developing it further the paper publishes the prediction that separates the two ephemerides (src/previsao_232634196.py; expectation written and committed before running). For cycle E = 2352 counted from the sector-19 primary minimum at BJD_TDB 2458828.9247 — the primary minimum nearest to 2028.0 — the quadratic of Table 3 predicts BJD_TDB 2461771.4651 ± 2.5 min (2027-12-31 23:09 TDB; full covariance, the cross term between the linear and quadratic coefficients included) and the linear ephemeris of the same seven epochs predicts 2461771.4792 ± 1.5 min (23:30 TDB; a formal bar, on a fit with reduced χ^2 21.7), 20.3 min later: 8.1σ of the quadratic prediction. One primary minimum timed to a minute in 2027–2028 — one TESS sector or one night of CCD photometry — decides between them, and a minimum that matches neither says the O–C is not polynomial, the third possibility this paper cannot exclude.

6. What is and is not claimed

Claimed:

- the coefficients of Table 3 and their distribution in the three partitions of Table 4 — sign-symmetric in each, with a median $|dP/dt|$ that depends on the partition — for the subpopulation defined by Tables 1–2, as quadratic coefficients over 16–20 yr and not as rates;
- that where an independent 34-yr record exists (CV Dra), the same window reproduces the coefficient and the whole record does not: the coefficient is local;
- that with formal timing errors the same chain would have reported 24 of 31 targets as significant period changes;
- that the data-derived bars, once the composition error of Sect. 5.2.2 is corrected, describe the residuals: median reduced χ^2 0.67 against 0.73 expected, aggregate 1.20 ($p = 0.13$);
- that the residual test with bars co-estimated from two or three samples has a null of 0.77, not 1, and that the apparent SuperWASP excess of 2.73 was the product of two estimation errors — a misplaced $\sqrt{n}$ (factor 1.74) and the small-sample bias of the estimator — and of three targets whose quadratic goodness-of-fit test rejects; that over the 23 targets with an adequate quadratic neither archive shows excess against the null with the same cut applied inside it (SuperWASP 0.73, 30th percentile of 0.77; TESS 0.55, 23rd percentile of 0.70), and that no bar correction is therefore applied — with the limit that the test does not separate correct bars from bars inflated by real secular terms of the tabulated size, which lower the factor in the same way (Sect. 5.2.3);
- that the ten detections are carried by structure within the five years of TESS, which survives the within-sector bars but not bars that include a between-sector drift of the size measured on the secondaries — a ceiling the TESS residuals neither require nor exclude — with the exception of TIC 232634196, which is carried by the 20-yr lever, is insensitive to the archival bars, has a secondary moving in the same direction as its primary with a common-mode change smaller than the differential one, and for which a falsifiable prediction is given (Sect. 5.5: quadratic and linear minima 20 min apart at 2028.0);
- that the completeness of the chain for a secular term, with the same estimator, is 50% at 0.015 s yr^{-1} (≥ 0.5 in 16 of 26 at 0.02 s yr^{-1} ; 0.54–1.00 at the tabulated coefficients of D10), and that the co-estimated bars suppress the significance of a real term by a factor that grows with it — median 1.15 at 0.02 and 2.8 at 0.2 s yr^{-1} , 5–24 in the nine targets with the smallest archival bars — leaving the coefficient unbiased and the goodness-of-fit test blind (Sect. 5.2.5);
- that a parabola fitted to an undersampled third-body orbit passes the significance, adversarial and goodness-of-fit tests used here, demonstrated on V527 Dra and quantified by injection; and that, with a per-target efficiency measured on a declared orbit population and the in-window tertiary fraction of Tokovinin et al. (2006), such cases are expected to be at most one or two of the ten (the efficiency is measured on fixed bars and is a ceiling for the pipeline’s own).

Not claimed:

- that any tabulated coefficient is secular. No member of D8 (or of D10) is externally confirmed as secular: six have no record long enough to test, and of the two that do, one fails (V527 Dra, published cyclic O–C) and the other agrees only within our window (V564 Dra, a different coefficient over 26 yr);

- the functional form of any O–C diagram; the presence of third bodies in any system beyond the published one; the physical mechanism behind any individual coefficient;
- that the adversarial test has a significance level: it is a robustness gate, a heuristic (Sect. 3.5);
- an informative bound on the in-window tertiary fraction from the residuals: the TESS side excludes only $f \gtrsim 0.8$, above every published value (Sect. 5.2.4);
- that the TESS bars inflated by 1.0 min yr^{-1} are calibrated (they are a ceiling);
- that the SuperWASP chain bias of $-2.14 \pm 0.78 \text{ min}$, calibrated on four hot Jupiters, transfers to deep, short eclipses — it is carried as a declared limitation, bounded only by two archival minima of CV Dra ($n = 2$) — the residual test of Sect. 5.2.3 is blind to a bias common to a target’s seasons, because the quadratic absorbs it; propagating its uncertainty as a common offset of the archival block rather than as per-epoch noise changes no verdict (Sect. 3.4);
- that the tabulated $\sigma(dP/dt)$ is the measurement error of a real secular term: for a term of $0.1\text{--}0.2 \text{ s yr}^{-1}$ on the designs with 1.5–5 min archival bars it is an overestimate by up to an order of magnitude (Sect. 5.2.5);
- that the timing of TIC 377253090 is free of its unresolved $1.7''$ companion of equal brightness: a constant companion does not move a minimum, but its variability was not tested (Sect. 3.1);
- that the nine TESS-internal detections are period changes rather than timing drift; nor that the bars of either archive are validated at the between-sector scale, which the data of this paper cannot test.

The flagged targets are not developed as case studies here. Earlier readings withdrawn in the course of the work are listed in Appendix D.

7. Data and code

Catalogues: Prša et al. (2022; VizieR J/ApJS/258/16), Kostov et al. (2025; J/ApJS/279/50). Photometry: TESS (Ricker et al. 2015) SPOC 2-min light curves (Jenkins et al. 2016) from MAST; SuperWASP public archive (Butters et al. 2010) as served by CERIT-SC (<https://wasp.cerit-sc.cz>). Published timings: VarAstro star catalogue and O–C data (var.astro.cz; Paschke & Brát 2006), queried 2026-09-13; bibliographies from SIMBAD (Wenger et al. 2000) and catalogues from VizieR (Ochsenbein et al. 2000), both queried 2026-09-13. Software: astropy (Astropy Collaboration 2013, 2018, 2022), numpy, scipy, pandas.

8. Data availability

The code that produces every number in this paper is archived at Zenodo under DOI [to be assigned at submission] and developed in a public repository at [URL to be inserted]; the archived version is the one this paper was generated from. It comprises the timing chain (`src/cadeia_oc.py`, `src/oc_lote.py`, `src/superwasp.py`, `src/epocas_142874476.py`), the measurement scripts of Sect. 5 (`src/literatura_26.py`, `src/comparar_brno_26.py`, `src/v527_ltte.py`, `src/secundarios_26.py`, `src/epsilon_26.py`, `src/reinflar_tess_26.py`, `src/qual_barra_26.py`, `src/nulo_barras_reestimadas_26.py`, `src/gls_vies_26.py`, `src/vies_alavanca_26.py`, `src/estrutura_swasp_26.py`, `src/caso_232634196.py`),

src/previsao_232634196.py, src/
1260 blending_26.py, src/completude_secular_26.py;
src/reinflar_swasp_26.py and src/
reinflar_conjunto_26.py are kept for the record of
the withdrawn corrections), the sector census of Sect. 2
(src/vies_cache_288.py), and the scripts that gen-
erate the tables and figures (src/tabela_notas.py,
src/figuras_notas.py, src/md_para_tex.py: Ta-
bles 3–5 and the figures are regenerated from the per-
target records, and the TeX from the manuscript source,
at every run). The archive also holds the per-target
1270 records (data/orquestra/oc_lote/oc__<TIC>.json),
the cross-match and comparison tables of
Sect. 5 (literatura_26, minimos_brno_26,
comparacao_brno_26, sobreposicao_brno_26,
v527_ltte_injetado, secundarios_26, epsilon_26,
reinflar_tess_26, qual_barra_26, reinflar_swasp_26,
reinflar_conjunto_26, vies_alavanca_26,
estrutura_swasp_26, estrutura_swasp_26_grade_f,
estrutura_swasp_26_reestimada,
nulo_barras_reestimadas_26,
1280 nulo_condicionado_26, gls_vies_26,
comparacao_estado_A_B, blending_crowdsap_26,
blending_gaia_26, completude_secular_26,
completude_secular_26_resumo,
completude_secular_26_alvo,
completude_secular_26_fator, all .parquet, and
caso_232634196.json, previsao_232634196.json, in
data/orquestra/oc_lote/), the MAST sector census of
the 288 targets (data/results/vies_cache_288.parquet),
the machine-readable table of the 145 epochs entering
1290 Table 3 (reports/epocas_26.csv, generated by src/
epocas_cds_26.py: TIC, source, cycle, BJD_TDB, assigned
and formal bars; to be deposited at the CDS), and the figures.
The TESS light curves are public at MAST, the SuperWASP
light curves at the CERIT-SC archive, and the VarAstro minima
at var.astro.cz.

Appendix A: Outcome of the recovery pass, per target

Of the eight targets whose period ladder had not closed (Table 2, row 4), intermediate 2-min sectors were fetched from MAST (96 light curves). Three of the eight also have SuperWASP curves below the 500-point floor and belong in row 1; one closed its ladder with the added sectors and then failed phase coverage (row 2); two have SuperWASP sources for which the archive returns an error page (retrieved four times); the remaining four — all in the northern continuous-viewing zone, with 22–27 sectors each once the MAST holdings were added — have individual sectors whose half-sector timing dispersion reaches 128 min, which the ladder cannot accommodate without an outlier-sector rule that does not yet exist.

Of the six without an applicable eclipse duration (row 5), one is a contact binary (morphology 0.99, catalogue width 36.6 h for $P = 4.8$ d) that the trapezoid does not describe and was not attempted. For the five others the eclipse width was measured on the folded TESS curve; for two of them the estimator returned exactly its 35%-of-period ceiling. That value is not a measurement of T_{14} — it is the estimator failing to locate a minimum when the curve is folded on the *catalogue* period, which points to the catalogue period rather than to the duration. These two are recorded as “minimum not found at the published period”, a different statement from “duration not published”, and one that would send a follow-up to the period, not to the eclipse. None of the six was recovered.

Appendix B: Control target and period-ladder audit

Before any new target was measured, the chain was run as a single function on TIC 142874476 and compared, component by component, with its previously recorded result — the four TESS epochs and their errors, the linear and quadratic residuals, χ^2 , Q , the SuperWASP source, cycle count, seasons, block-coverage status, the 2006–2008 epoch and its error, and the final residuals. Thirty of 38 components reproduced to the quoted precision. The eight that did not were all downstream of one number — a quadratic extrapolation to 2007 of +38.9 min that existed only in a comment and that no fitting convention reproduces (weighted: +28.2; unweighted: +28.7; $Q \cdot E^2$ alone: +43.5). The record was corrected to what the code produces: the 2007 epoch is 10.2σ from the linear ephemeris and 2.4σ from the quadratic; the earlier statement that a pure parabola was rejected at 7.9σ does not survive, and no claim about the form of that O–C is made here. The adversarial test on the control gives $p = 0.28$ for the TESS-internal curvature ($p = 0.003$ before the shift); with four epochs and one degree of freedom, that detection is fragile, which is why the robust statement about the control is the 10σ departure from a constant period, not the curvature. The control was re-run before and after every change reported in this version and reproduced each time.

Period ladder. The ladder returns epochs and residuals in the caller’s order. This was a latent defect: an earlier version returned them in internal time order, and unsorted input produced cycle numbers assigned to the wrong sectors — caught only because the resulting dP/dt of $-24\,122\text{ s yr}^{-1}$ is physically impossible; a milder scramble would have looked like mass transfer. The $|dP/dt| > 10\text{ s yr}^{-1}$ guard of Sect. 3.2 exists because of it. The tolerance rule of Sect. 3.2 (the larger of 3 min and $3\times$ the median epoch uncertainty) was introduced after the main batch, which ran with a fixed 3 min; re-running all 26 targets under the rule

reproduces every tabulated value (1 024 fields compared across the 26 per-target records; only the wall-clock time differs).

Appendix C: The orbit population, the efficiency, and the occurrence numbers

Population. Circular tertiary orbits with P_3 log-uniform in 0.1–20 yr (between a sector and the baseline), binary mass uniform in $1.5\text{--}2.5\,M_\odot$, tertiary mass uniform in $0.1\text{--}1.0\,M_\odot$, isotropic inclination ($\cos i$ uniform), phase uniform; the light-travel-time semi-amplitude follows from Kepler’s third law, $A = a_3\,M_3/(M_{\text{bin}} + M_3) \sin i / c$, and reproduces the published 6.6 min of V527 Dra for a $1\,M_\odot$ tertiary at 70° and $P_3 = 2.733$ yr. The median amplitude of the population is 2.0 min, comparable to the epoch bars. ε_3 (detection and $p_{\text{gof}} \geq 0.05$) sums to 4.1.

What sets ε . The expectation that ε would track the sampling pattern — fewer epoch groups, higher ε , the V527 Dra mechanism — was not met: the Spearman correlation of ε with the number of epoch groups is +0.09 ($p = 0.66$) and with N is +0.31 ($p = 0.13$), while with the median bar it is -0.64 . Two targets (TIC 224605072, 229476285) have $\varepsilon = 0$: their SuperWASP bars make the curvature test powerless at these amplitudes, so their “no significant curvature” is uninformative and they are excluded from the sensitivity-restricted row of Table 4. ε depends on the declared population; for orbits as favourable as V527 Dra’s it approaches 0.83. The one empirical anchor points to that ceiling — V527 Dra’s amplitude is three times the population median — and is biased: the orbit was published because it is large enough to solve from 1 832 timings, so the only known triple being a favourable one is what publication selection produces, and with $n = 1$ it does not inform the population.

Occurrence numbers and what each measures. Occurrence over all outer periods: 96% of solar-type spectroscopic binaries with $P < 3$ d after correction for incompleteness, 79% raw (Tokovinin et al. 2006); a lower limit of $42 \pm 5\%$ for contact binaries (Pribulla & Rucinski 2006). Detected fractions in eclipse-timing surveys: 8.5% of $\sim 2\,600$ Kepler binaries with outer periods below $\sim 2\,000$ d (Borkovits et al. 2016); 1.2% of $\sim 80\,000$ OGLE-IV binaries (Hajdu et al. 2019); 10 in ~ 100 TESS full-frame-image targets (Marcadon & Prša 2024). The detected fractions are convolved with each survey’s baseline, cadence and precision; multiplying them by ε applies an efficiency twice. The in-window fraction of Sect. 5.4 comes from the 33 tertiaries listed for the 42 $P < 3$ d systems of Tokovinin et al.’s Table 5, all ten in-window ones being spectroscopic or visual-orbit detections rather than resolved wide pairs.

Appendix D: Record of readings withdrawn between versions

Each entry names the reading, the version that carried it, and what replaced it. They are kept here rather than in the body because the body is about the data; they are kept at all because each was, at the time, the honest reading of a number that turned out to measure something else. They fall into three groups.

Errors in the estimator or in what it was compared with.

1. A median reduced χ^2 of 1.41 read as bars 1.39 times too small, and the archival variance factor of 2.73 read as SuperWASP bars 1.65 times too small, with a “corrected” set of ten detections and a “conservative joint correction” (SuperWASP $\times 1.65$, TESS $\times 1.34$) leaving eight. The composition error of Sect. 5.2.2 — the standard error of the season mean

assigned to each season as its bar — accounted for a factor of 1.74; corrected, the factor is 1.35 and the median 0.67, and the corrections are withdrawn.

2. *The remaining excess as small-sample bias.* The expectation written for the null test of Sect. 5.2.3 was a SuperWASP null of 1.3–1.8 from the $\nu/(\nu - 2)$ argument; the null is 0.77, because r and s are the same numbers, and the measured 1.35 is an excess located in the three targets with a rejected quadratic.
3. *The 23 adequate targets against an unconditioned null.* A version compared the residuals of the 23 targets that passed the goodness-of-fit cut with a null simulated on the same 23 designs without the cut (0.78; TESS 0.96, the measured 0.55 at its 13th percentile). With the cut applied inside the null the TESS null is 0.70 and the measured value is at the 23rd percentile (Sect. 5.2.3); the same conditioning was propagated to the injection of Sect. 5.2.4, where the 23 had likewise been compared without the cut.
4. *A TESS-side bound of $f \lesssim 0.5$ on the in-window tertiary fraction,* and the reading that orbits at $f = 0.24$ explain 42% of the archival excess, both obtained by injecting with the bars held fixed and comparing with a measurement whose estimator re-estimates them. Replaced by the injection with the matched estimator, whose TESS-side bound ($f \gtrsim 0.8$ excluded) is uninformative, and an archival side nearly blind to orbits (Sect. 5.2.4).

Transfers from one object or one population to another.

1. *A borrowed efficiency.* A version multiplied survey-detected tertiary fractions by an efficiency of 0.83 taken from the V527 Dra injection; the fractions are already convolved with each survey's efficiency, and the efficiency of one orbit at one sampling pattern does not transfer. Replaced by the per-target ε on a declared population and the in-window fraction of Tokovinin et al. (Sect. 5.4).
2. *“One in nine verified”.* A version quoted a false-positive rate of one in nine against the literature; seven of the nine had no record long enough to test. Replaced by an existence proof and an injection rate (Sect. 5.4).
3. *The residuals as a bound on the transferred chain bias.* A version stated that the SuperWASP residuals of the 23 adequate targets, showing no excess, left no room for a shape-dependent bias larger than the bars. A bias common to a target's seasons is a block offset that the quadratic absorbs; the residual test cannot see it, and the only bound is the two CV Dra minima of Sect. 5.5 (Sect. 3.4).

Criteria that measured something other than what they were named for.

1. *A fixed flag $\chi^2_{\text{red}} \geq 2$ as a validation.* An early version flagged 7 of 26 targets by a fixed reduced χ^2 and read the flag on V527 Dra as a blind validation; the flag's false-alarm probability depends on the degrees of freedom (0.157 at 1, 0.092 at 4), a coincidence with 7 of 26 flagged had a 27% prior probability, and the criterion was replaced by $p_{\text{gof}} < 0.05$ (Sect. 3.5).
2. *TESS bars “calibrated” at 1.0 min yr^{-1} .* A version called the TESS bars inflated by the secondary drift calibrated because they brought the median reduced χ^2 to 0.73; the coincidence came from re-weighting, and the inflation is a ceiling (Sect. 5.3.2).

3. *Eleven detections and nine adequate ones.* TIC 359552377 left the set when the composition error was corrected (Sect. 4; $p_{\text{adv}} 0.05 \rightarrow 0.22$). Its departure has two readings that the data do not separate: a spurious detection removed by the corrected bars, or a real term of the size it carries ($-0.25 \pm 0.08 \text{ syr}^{-1}$) that the gate on this design — five epochs, archival bars of 16 min — passes in 26% of injections (Sect. 5.2.5). The count is the same under both; what it means is not.

Acknowledgements. An external reader of an earlier version asked whether the archival variance factor could be an artefact of its denominator; the literal check of that question found the composition error of Sect. 5.2.2 and led to the null test of Sect. 5.2.3, and the paper owes its central section to it. This research has made use of the SIMBAD database and the Vizier catalogue access tool, operated at CDS, Strasbourg, France; of the VarAstro database of the Variable Star and Exoplanet Section of the Czech Astronomical Society; of data from the TESS mission obtained from the Mikulski Archive for Space Telescopes (MAST), funded by the NASA Explorer Program; and of the SuperWASP public archive.

References

- Astropy Collaboration, Robitaille, T. P., Tollerud, E. J., et al. 2013, A&A, 558, A33, doi:10.1051/0004-6361/201322068
- Astropy Collaboration, Price-Whelan, A. M., Sipőcz, B. M., et al. 2018, AJ, 156, 123, doi:10.3847/1538-3881/aabc4f
- Astropy Collaboration, Price-Whelan, A. M., Lim, P. L., et al. 2022, ApJ, 935, 167, doi:10.3847/1538-4357/ac7c74
- Borkovits, T., Hajdu, T., Sztakovics, J., et al. 2016, MNRAS, 455, 4136, doi:10.1093/mnras/stv2530
- Butters, O. W., West, R. G., Anderson, D. R., et al. 2010, A&A, 520, L10, doi:10.1051/0004-6361/201015655
- Čeki, A., Şenavcı, H. V., Latković, O., Uzunçam, E., Yorulmaz, E. B., & Bahar, E. 2024, MNRAS, 532, 3582, doi:10.1093/mnras/stae1709
- Hajdu, T., Borkovits, T., Forgács-Dajka, E., et al. 2019, MNRAS, 485, 2562, doi:10.1093/mnras/stz592
- Jenkins, J. M., Twicken, J. D., McCauliff, S., et al. 2016, Proc. SPIE, 9913, 99133E, doi:10.1117/12.2233418
- Kostov, V. B., Powell, B. P., Fornear, A. U., et al. 2025, ApJS, 279, 50, doi:10.3847/1538-4365/ade2d8
- Marcadon, F., & Prša, A. 2024, ApJ, 976, 242, doi:10.3847/1538-4357/ad8571
- Ochsenbein, F., Bauer, P., & Marcout, J. 2000, A&AS, 143, 23
- Paschke, A., & Brát, L. 2006, OEJV, 23, 13
- Pawar, T., Miszuda, A., Helminiak, K. G., et al. 2026, A&A, 705, A251, doi:10.1051/0004-6361/202554310 (arXiv:2502.20626)
- Pribulla, T., & Rucinski, S. M. 2006, AJ, 131, 2986, doi:10.1086/503871
- Prša, A., Kochoska, A., Conroy, K. E., et al. 2022, ApJS, 258, 16, doi:10.3847/1538-4365/ac324a
- Pyatnytsky, M. 2020, JAAVSO, 48, 171
- Ricker, G. R., Winn, J. N., Vanderspek, R., et al. 2015, JATIS, 1, 014003, doi:10.1117/1.JATIS.1.1.014003
- Tanriver, M., Poro, A., Bulut, A., Keskin, A., & Blackford, M. G. 2023, RAA, 23, 055005, doi:10.1088/1674-4527/acc506 (arXiv:2303.09109)
- Tokovinin, A., Thomas, S., Sterzik, M., & Udry, S. 2006, A&A, 450, 681, doi:10.1051/0004-6361:20054427
- Wenger, M., Ochsenbein, F., Egret, D., et al. 2000, A&AS, 143, 9